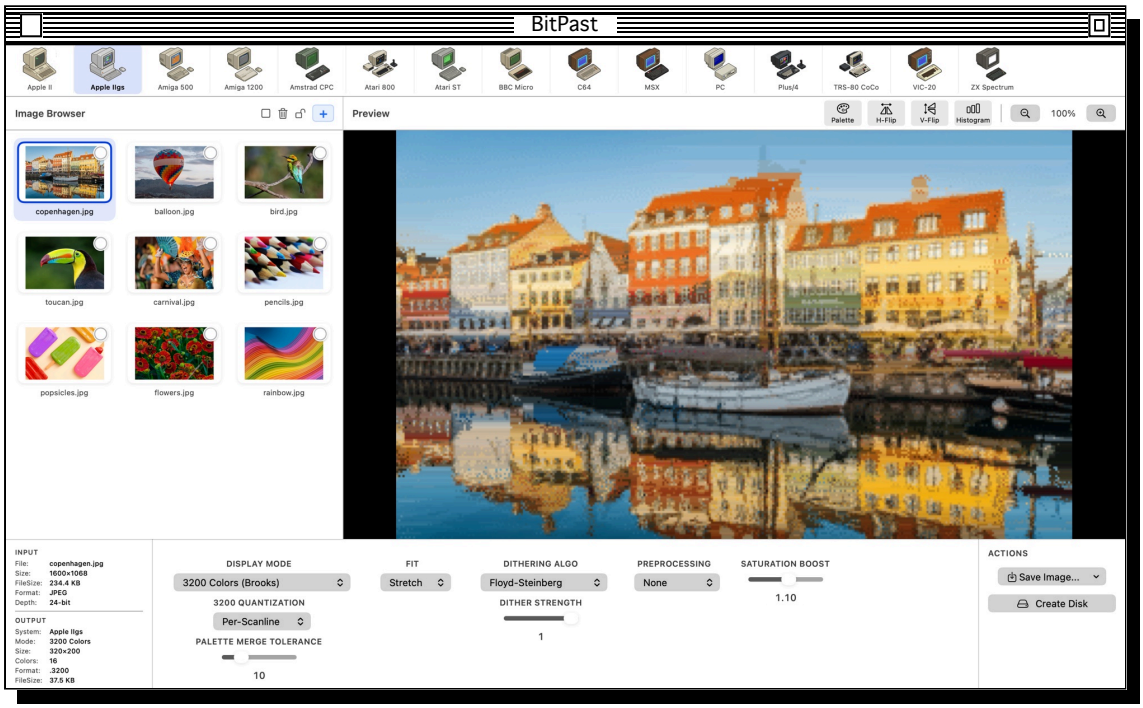
FOR MACOS

BitPast

User's Guide and Technical Reference

Turn modern pictures into graphics for fifteen classic computers



VERSION 4.6

BitPast

User's Guide and Technical Reference

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Written by Walter Tengler.

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The sample photographs are from Pixabay and are used under the Pixabay Content License; see Appendix D for the photographers. The pictures of the old computers' graphics were made by BitPast itself.

Every effort has been made to make this guide accurate. The settings, modes and file sizes it lists were taken from BitPast version 4.6 itself.

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About This Guide

This guide explains how to use BitPast, and lists what it can do: every system it supports, every graphics mode and every setting. It is meant to be read from the beginning, and used afterwards as a reference.

If you are new

Read Chapters 1 to 3, then try the exercise in Chapter 2. The rest can wait until you need it.

If you want a system

Chapter 6 has a section for each of the fifteen systems, with every mode, its picture, its file and its settings. Appendix A lists the files; Appendix B shows the colour palettes.

Conventions

Names you see on the screen — menu items, buttons and pop-up menus — are set in this typeface: `Create Disk`. File names and extensions look like this: `.shr`. Keyboard shortcuts are shown with symbols: ⇧⌘1 means hold Shift and Command and press 1.

Three kinds of note appear in the margin:

NOTE A **Note** adds something that is good to know.

IMPORTANT An **Important** note warns you about something that could lose work or surprise you.

TIP A **Tip** is a suggestion that makes the result better or the work quicker.

CHAPTER 1

1 Welcome to BitPast

Turn today's pictures into the graphics of yesterday's computers

BitPast converts ordinary pictures — photographs, drawings, screenshots — into graphics for fifteen classic home computers. It does not simply shrink a picture and reduce its colours. Each conversion follows the rules of the real hardware.

A ZX Spectrum picture may use only two colours in every 8×8 cell. A Commodore 64 multicolor picture has four colours in each 4×8 cell. An Apple IIgs picture can use a different palette on every line of the screen. BitPast makes these choices for you, shows the result at once, and saves it in a file that the machine or an emulator can load.

1.1 What BitPast does

Convert

Add pictures, choose a system, and the preview shows how the picture will look on that machine. Every change to a setting converts the picture again.

Save

Save the preview as PNG, JPEG, GIF or TIFF to share it, or save the machine's native file — a `.shr` for the Apple IIgs, a `.kla` for the Commodore 64, a `.scr` for the ZX Spectrum — to load on the real thing.

Disk

Put a whole set of converted pictures onto a disk image for the system — `.d64`, `.adf`, `.po`, `.atr` and more — that an emulator, a floppy emulator or a disk-copying tool can use.



FIGURE 1-1
One photograph, fifteen machines. Each picture was converted in the system's default mode, with default settings.

1.2 How a conversion works

Every conversion passes the picture through the same six stages. Knowing them helps you decide which setting to change when a result is not what you hoped for.

STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5	STAGE 6
Picture	**Place**	**Prepare**	**Colours**	**Dither**	**Save**
Any picture macOS can open: JPEG, PNG, GIF, TIFF, BMP, HEIC and more.	Fit, Aspect and the position decide how the picture fills the screen.	Filters, contrast, saturation and gamma make the picture suit a small palette.	BitPast chooses the best colours the hardware allows, often for each cell or line.	Patterns of dots fake the colours in between.	A preview picture, the native file, or a whole disk image.

Conversions run in the background, so the window stays responsive while a picture is worked out. When a picture takes a moment, a small Converting… indicator appears over the preview.

Only the preview is a picture. The native file contains the screen exactly as the machine's hardware stores it — bitmaps, colour attributes, palettes — in the layout the machine expects.

1.3 What you need

A Mac

BitPast needs macOS 15.7 or later.

Pictures Any picture macOS can read. Pictures with strong shapes and clear colours convert best, especially for the systems with the smallest palettes.

Somewhere to try the files An emulator for the system you chose, or the real machine with a way to get files onto it. The previews need nothing.

Where to get it BitPast is available from the Mac App Store. Releases are also published at github.com/portwally/BitPast.

1.4 The fifteen systems at a glance

SYSTEM	INTRODUCED	SCREEN SIZES	COLOURS	NATIVE FILES
Apple II	1977	280×192, 560×192; Lo-Res 40×48, 80×48	6 or 16	`.BIN .A2FC .A2FM .SLO .DLO`
Apple IIgs	1986	320×200, 640×200	up to 3,200	`.SHR .3200`
Amiga 500	1987	320×256, 320×512	32 of 4,096; HAM6	`.iff`
Amiga 1200	1992	320×256 to 640×512	256; HAM8 262,144	`.iff`
Amstrad CPC	1984	160×200, 320×200	16 or 4 of 27	`.scr`
Atari 800	1979	80 to 320 × 192	2 to 16 of 128	`.gr8 .gr9 .gr10 .gr11 .gr15`
Atari ST	1985	320×200, 640×200, 640×400	16, 4, 2	`.pi1 .pi2 .pi3`
BBC Micro	1981	160 to 640 × 256	2, 4 or 8	`.bbc`
Commodore 64	1982	160×200, 320×200, 40×25	16	`.kla .art .prg`
MSX	1983	256×192, 256×212	16, 16 of 512, 256	`.sc2 .sc5 .sc8`
IBM PC	1981	320×200; text 80×25, 132×50	4, 16, 256	`.pcx .ans`
Commodore Plus/4	1984	160×200, 320×200	128	`.prg`
TRS-80 CoCo	1980	128×96 to 640×200	2, 4 or 16	`.bin .cm3`
VIC-20	1980	88×184, 176×184	16	`.prg`
ZX Spectrum	1982	256×192	15	`.scr`

CHAPTER 2

2 Getting Started

From installation to your first saved picture

This chapter takes you from installing BitPast to saving your first picture. It takes about five minutes.

2.1 Installing BitPast

From the App Store

Open the Mac App Store, search for BitPast and click Get. BitPast appears in your Applications folder.

From GitHub

Download `BitPast.zip` from the Releases page at github.com/portwally/BitPast, open it, and drag BitPast into your Applications folder.

NOTE

A copy downloaded from GitHub may not be notarized by Apple, so macOS can show a warning the first time: the app "cannot be opened because it is from an unverified developer." Close the warning, then Control-click the BitPast icon in the Applications folder, choose Open from the menu and click Open again. macOS remembers your decision.

If that does not work, open Terminal and type `xattr -cr /Applications/BitPast.app`, then try again.

2.2 Your first conversion

1. **Open BitPast.** The window is empty: the Image Browser on the left says "Drag Images Here".
2. **Add pictures.** Drag one or more pictures from the Finder into the Image Browser, or click the + button above it and choose files. Each picture appears as a thumbnail.
3. **Select a picture.** Click a thumbnail. A blue frame marks the picture you are working on, and the preview shows its conversion.
4. **Choose a system.** Click one of the fifteen icons across the top of the window — for example Apple IIgs. The preview changes at once.
5. **Choose a mode and adjust.** The panel at the bottom holds the settings for the system. Try the Display Mode pop-up menu, then Dithering Algo. Each change converts the picture again.
6. **Save.** Click Save Image… in the Actions column. Choose PNG for a picture you can share, or Native Format for the file the machine loads.

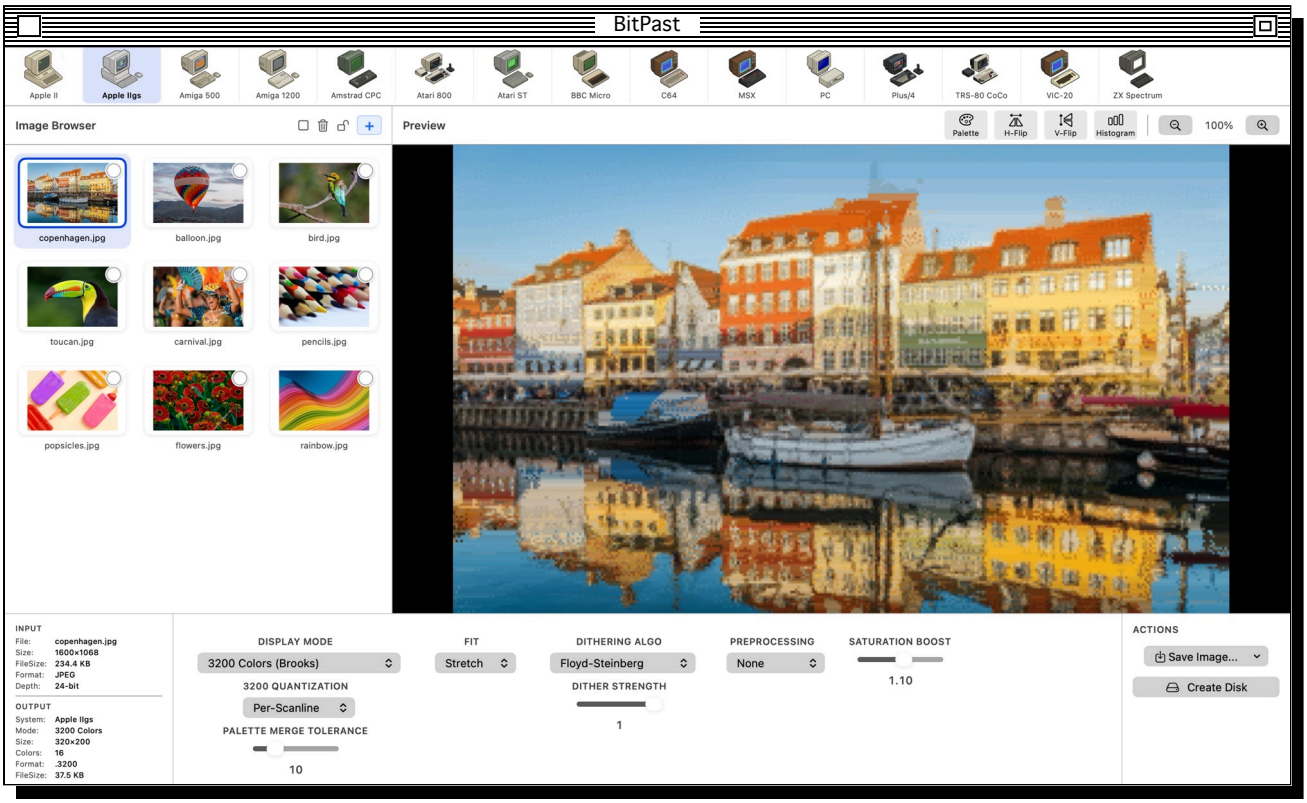


FIGURE 2-1
BitPast with a picture selected and the Apple IIgs chosen.

2.3 What you just made

The picture in the preview is what the machine would show on its screen, scaled up. The panel at the bottom left, headed Output, tells you what the file contains: the system, the mode, the size in pixels, the number of colours, the file format and the file size.

The file you saved with Native Format is not an ordinary picture. It is the screen as the machine stores it, and only the machine, an emulator or a viewer for that system can show it. The PNG is an ordinary picture you can open anywhere.

TIP Convert the same picture for several systems, one after another. The picture stays selected when you change system, and each system remembers its own settings while BitPast is open. This is the quickest way to learn what each machine does to a picture.

CHAPTER 3

3 The BitPast Window

A tour of everything you see

BitPast has one main window. This chapter names its parts, and the chapters that follow refer to them.

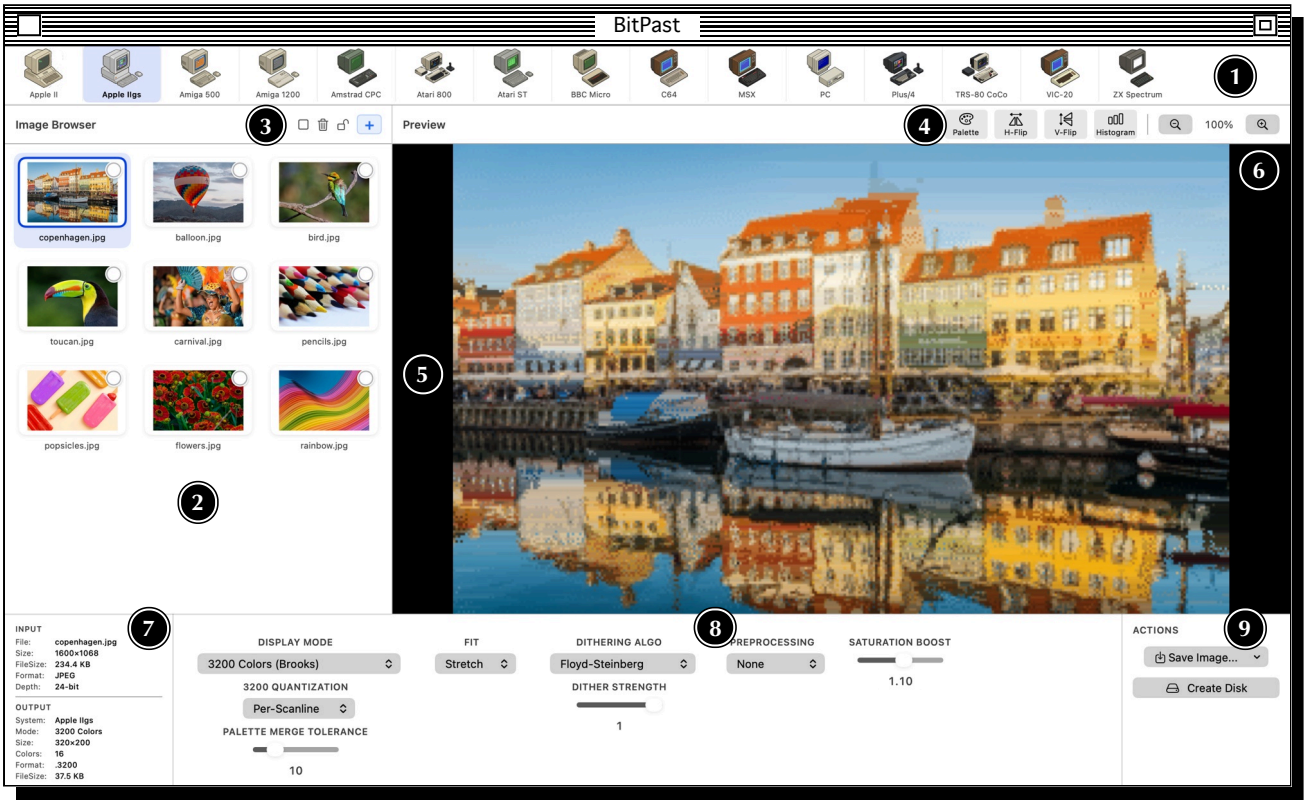


FIGURE 3-1
The parts of the BitPast window.

1. **System bar.** Fifteen icons, one for each system.
2. **Image Browser.** The pictures you have added.
3. **Image Browser tools.** Select all, remove, lock settings, add.
4. **Preview tools.** Palette editor (Apple IIgs only), flip, histogram.
5. **Preview.** The converted picture.
6. **Zoom.** Magnify or reduce the preview.
7. **Info panel.** Facts about the picture and the result.
8. **Settings panel.** The controls for the selected system.
9. **Actions.** Save the picture, or create a disk image.

3.1 The system bar

The bar across the top shows an icon for each of the fifteen systems. Click one to convert for it. The selected system is highlighted, and the settings panel, the preview and the Output facts change to match.

Each system keeps its own settings while BitPast is open, so you can move between systems without losing what you set. Pictures with locked settings (Chapter 4) are released when you change system.

Keyboard

The System menu lists the same fifteen systems, each with a shortcut: ⇧⌘1 to ⇧⌘0 for the first ten, then ⇧⌘P, ⇧⌘L, ⇧⌘T, ⇧⌘V and ⇧⌘Z. Chapter 10 has the full list.

3.2 The Image Browser

The browser on the left holds every picture you have added, as thumbnails with their file names. The picture with the blue frame is the one shown in the preview. The small circle at the top right of each thumbnail selects it for batch work.

Four tools sit above the thumbnails: a checkbox that selects or deselects every picture, a trash can that removes the selected pictures, a padlock that locks the current settings for the selected picture, and +, which adds pictures.

3.3 The preview

The preview shows the conversion of the selected picture. Above it, from left to right: Palette (Apple IIgs only), H-Flip, V-Flip and Histogram, then the zoom buttons. While a new conversion is worked out, a Converting… indicator appears over the old picture.

When Fit is set to something other than Stretch, you can drag the preview to choose which part of the picture shows. Chapter 4 explains how.

3.4 The info panel

The panel at the bottom left tells you about the picture and the result. Under Input: the file name, the size in pixels, the file size, the format and the colour depth. Under Output: the system, the mode, the size, the number of colours, the file format and the file size of the native file.

3.5 The settings panel

The wide panel in the middle holds the settings for the selected system. They are arranged in groups: the mode first, then dithering, then contrast and filters, then colour. The list is different for every system, because every machine has its own choices; Chapters 5 and 6 describe them all.

Some controls appear only when they matter. For example, SCB Mode shows only in the Apple IIgs's 256-colour mode, and Aspect shows only when Fit is not Stretch.

3.6 The Actions column

Save Image… is a menu. It saves the preview as PNG, JPG, GIF or TIFF, or the machine's file as Native Format. When several pictures are selected it changes to Save 3 Images… (or however many you selected) and saves them all.

Create Disk opens a sheet for making a disk image of the system's files (Chapter 9).

3.7 The menus

System

Switch to any of the fifteen systems, with a shortcut for each.

Help

Graphics Mode Guide (⌘?) opens a window describing every system, mode and setting — the same information as this guide, inside the app. Visit GitHub Page opens the project page in your browser.

Settings

Choose BitPast ▸ Settings… (⌘,) to change the appearance of the app (Chapter 10).

CHAPTER 4

4 Working with Pictures

Adding, selecting, flipping, framing and locking

This chapter covers everything you can do with a picture before and after it is converted: adding and selecting pictures, looking at the result closely, and choosing how the picture fills the screen.

4.1 Adding pictures

Drag pictures from the Finder into the Image Browser. You can also click + above the browser and choose files. Add as many as you like: they appear as thumbnails, and the first picture you add is selected automatically.

BitPast opens any picture format macOS can read, such as JPEG, PNG, GIF, TIFF, BMP and HEIC. Adding a picture never changes the original file.

4.2 Selecting and removing pictures

Click a thumbnail to show it in the preview. That is the picture you are working on, and it has a blue frame.

To work on several pictures at once — to save them together, or to put them on a disk — use the small circle at the top right of each thumbnail. Click a circle to select that picture. Hold Shift and click another thumbnail to select every picture between the two. The checkbox above the browser selects or deselects all of them.

Remove

The trash can above the browser removes the selected pictures from BitPast. It does not delete the files from your Mac.

4.3 Looking closely

Zoom

The magnifying-glass buttons at the right end of the preview toolbar change the size of the preview. At 100% the picture is fitted to the space available. The buttons step the magnification by 20% at a time, up to about 500%. When the picture is larger than the preview, scroll to move around it.

Flip

H-Flip mirrors the picture left to right and V-Flip top to bottom. The flip is applied to the original picture before it is converted, so the saved file and any disk image are flipped too — not only the preview. A flipped picture's button stays lit; click it again

to undo the flip. The flip belongs to the picture, so other pictures are not affected, and it stays when you change settings or system.

Histogram

Histogram shows how the red, green and blue values (and, if you wish, the brightness) of the converted picture are spread. It is a good way to see how much of the range a small palette really uses. Click the letters R, G, B and L to show or hide a channel; close the panel with the × at its corner.

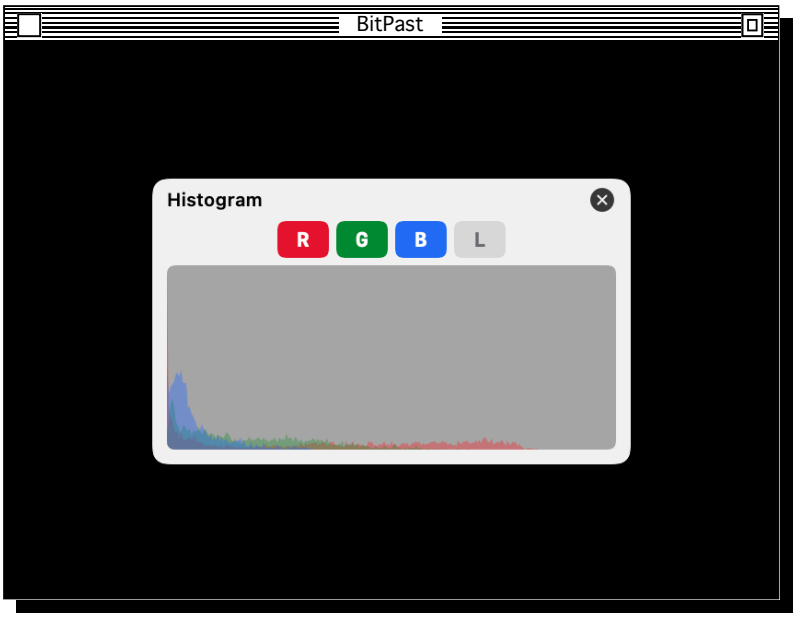


FIGURE 4-1
The histogram of a converted picture.
R, G and B show the colour channels; L shows brightness.

4.4 Fit, Aspect and position

Your picture and the machine's screen rarely have the same shape. **Fit** decides what happens when they differ. It is available for every system.

FIT	WHAT IT DOES
Stretch	Fills the screen and ignores the shape of the picture. This is the default, and the only choice before BitPast 4.6.
Contain	Shows the whole picture, with black bars where it does not fill the screen.
Cover	Fills the screen and crops whatever sticks out.
None	Does not scale at all: one picture pixel for each screen pixel, centred, so you see only the middle part of a large picture.



Stretch
The picture is squeezed to fit.



Contain
The whole picture, with bars at the sides.



Cover
The screen is filled; the top and bottom are cropped.



None
One picture pixel per screen pixel.

FIGURE 4-2
The four choices of Fit, with a tall picture on the Amiga 1200.

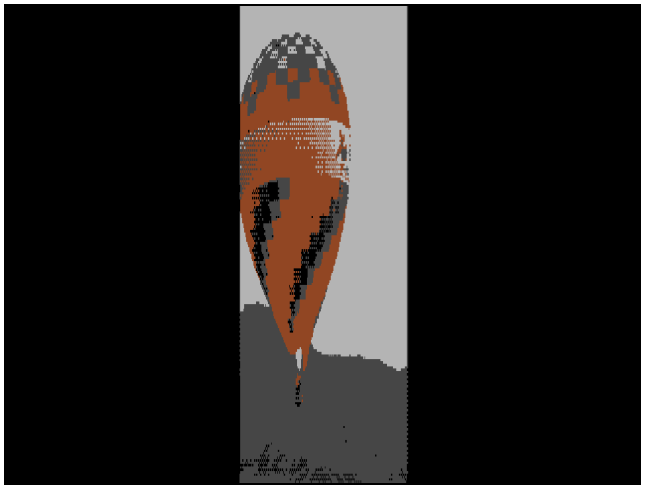
Aspect

The pixels of old computers are not square. On an Atari ST in medium resolution, for example, each pixel is much taller than it is wide, because 640 pixels share a screen with only 200 lines. **Aspect** takes this into account whenever Fit is not Stretch.

With **Correct**, BitPast assumes the picture is shown on a 4:3 screen and shapes it so circles stay round. With **Ignore**, it treats the pixels as square. The Aspect control appears only when Fit is Contain, Cover or None.



Correct
The proportions are right on the real screen.



Ignore
Pixels are treated as square, so on the real screen the picture comes out too narrow.

FIGURE 4-3
Aspect on the Atari ST in medium resolution, with Fit set to Contain.

Choosing which part shows

When Fit is Contain, Cover or None, you can move the picture. Point at the preview: the cursor turns into an open hand. **Drag** the preview and the picture follows your mouse; when you let go, BitPast converts it again at the new position. **Double-click** the preview to centre the picture again.

Each picture keeps its own position, and the position is used when you save, when you export several pictures and when you make a disk image. With Stretch there is nothing to move, so dragging does nothing.



Top
Dragged down.



Centre
The default.



Bottom
Dragged up.

FIGURE 4-4
The same picture with Fit set to Cover, at three positions.

IMPORTANT

Moving a picture converts it again, and so does every other change of settings. On the Apple IIgs, changes made in the palette editor are lost when the picture is converted again, so make them last (Chapter 7).

4.5 Locking settings for one picture

Normally one set of settings applies to every picture. When you want one picture to keep its own settings — a dark photograph that needs more contrast than the others, say — you can **lock** them.

1. Select the picture and adjust the settings until the preview looks right.
2. Click the padlock above the Image Browser. It turns orange, and the picture keeps these settings.
3. Select another picture and adjust the settings for it; lock that one too if you wish.
4. Save or export. Each locked picture is converted with its own settings; pictures that are not locked use the current settings.

To release a picture, select it and click the orange padlock again. Locked settings belong to the system you were using when you locked them: they apply only to that system, and they are released when you switch to another.

NOTE

A locked picture does not follow the settings panel. If a slider seems to do nothing, check whether the selected picture is locked.

4.6 Choosing and preparing pictures

- **Strong shapes win.** The smaller the palette, the more a picture needs clear shapes and contrast. A toucan on a dark background converts better than a distant landscape.
- **Crop first, or use Cover.** A close crop puts more of the machine's few pixels on the subject.
- **Contrast and saturation are cheap fixes.** A flat picture gets muddy in a small palette: try CLAHE contrast and a little extra saturation before anything else.
- **Size does not matter.** BitPast scales every picture to the machine's screen, so a large original gives a better result than a small one, never a worse one.
- **Compare systems.** Converting the same picture for several machines shows quickly which suits it.

CHAPTER 5

5 Settings Reference

The controls that every system shares

Most settings have the same name and the same effect on every system. This chapter describes them once. Settings that belong to one system only — the Apple II's b2d options, the IIgs's SCB Mode, the CoCo's colour sets — are described in the section for that system in Chapter 6.

5.1 Which settings each system has

SETTING	II	GS	A5	A12	CPC	800	ST	BBC	C64	MSX	PC	P/4	COCO	V20	ZX
Dithering	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Dither amount		•	•	•	•	•	•	•	•	•	•	•	•	•	•
Contrast			•	•	•	•	•	•	•	•	•	•	•	•	•
Filter			•	•	•	•	•	•	•	•	•	•	•	•	•
Colour match			•	•	•	•	•	•	•	•	•	•	•	•	•
Pixel merge					•				•			•		•	
Saturation		•	•	•	•	•	•	•	•	•	•	•	•	•	•
Gamma			•	•	•	•	•	•	•	•	•	•	•	•	•
Preprocessing		•													

● means the system offers the setting. Fit and Aspect (Chapter 4) are offered by every system.

5.2 Dithering

A machine with few colours cannot show a smooth gradient, so it fakes one with *dithering*: it mixes dots of the colours it has, and your eye blends them. There are two families of method.

Error diffusion (Floyd-Steinberg, Atkinson, Jarvis-Judice-Ninke, Stucki, Burkes and others) works through the picture dot by dot. Each time it must round a colour, it spreads the error over the dots that are still to come. The result looks organic, a little like grain. **Ordered dithering** (Bayer, Blue Noise) lays a fixed pattern of thresholds over the picture. It is more regular, and in the machines that work in cells it often looks cleaner.



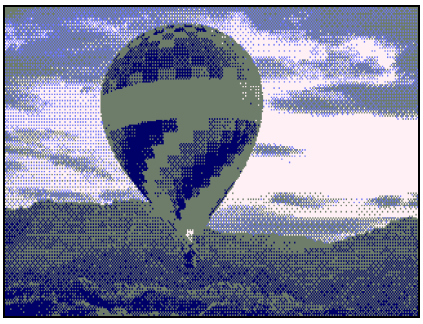
None
Each dot takes the nearest colour.



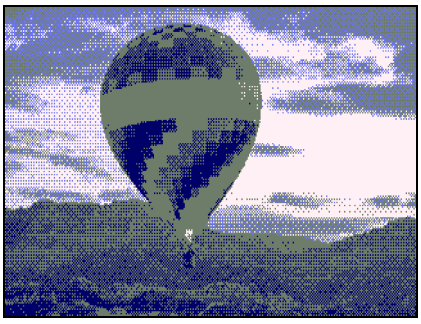
Floyd-Steinberg
The classic all-round choice.



Atkinson
Less diffusion; crisper, with a Macintosh look.



Bayer 4×4
A small, regular pattern.



Bayer 8×8
A larger pattern, smoother gradients.



Blue Noise 16×16
Irregular, but without clumps.

FIGURE 5-1
Six dithers on the Amstrad CPC in Mode 1, which has only four colours.

ALGORITHM	CHARACTER
None	No dithering; every dot takes the nearest colour.
Floyd-Steinberg	Classic error diffusion. A good all-round choice, and the usual one for photographs.
Atkinson	Diffuses less of the error, so highlights and shadows stay clean. It gives the classic early-Macintosh look.
Jarvis-Judice-Ninke	Spreads the error further; smoother gradients.
Stucki	Similar to Jarvis-Judice-Ninke, with a slightly different character.
Burkes	A faster variant of Stucki.
Noise	Random dither. Adds a film-grain texture. Because it is random, the same picture converts a little differently each time.
Bayer 2×2 to 16×16	Ordered pattern dither. Small sizes give a fine pattern; larger sizes give more levels but a more visible pattern.
Blue Noise 8×8, 16×16	Random-looking, but without clumps, so the texture is pleasing.

Not every system offers every method. The lists are in the settings tables in Chapter 6. The Apple II is converted by b2d, which has its own set of error-diffusion methods.

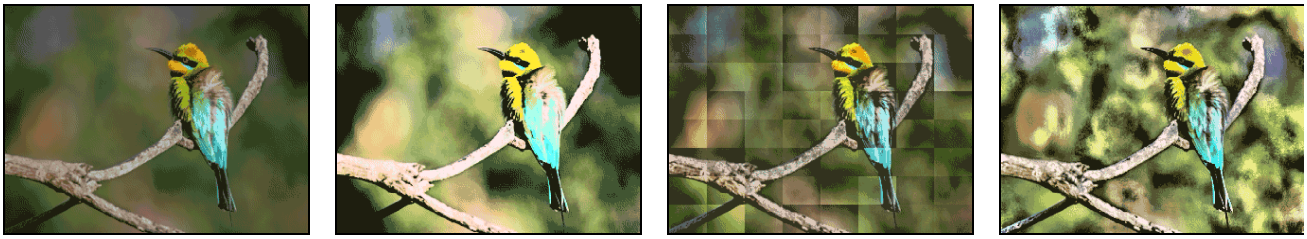
5.3 Dither amount

Dither Amount (called **Dither Strength** on the Apple IIgs) controls how strongly the dither is applied. 0 turns it off; 1 is full strength. Around 0.5 is a subtle dither and is the default for most systems. Values above 1 can produce artefacts.

5.4 Contrast

Contrast enhancement stretches the tones of the picture before the colours are chosen, which helps flat, dull pictures. Three methods are offered:

METHOD	WHAT IT DOES
None	No adjustment.
HE	Histogram Equalization: a global contrast stretch.
CLAHE	Local adaptive contrast. Good for photographs.
SWAHE	Sliding-window adaptive contrast. Smooth results.



None **HE** **CLAHE** **SWAHE**

FIGURE 5-2
The four contrast methods on the Amiga 500.

5.5 Filters and preprocessing

Most systems have a **Filter** menu with four quick filters: **Lowpass** (a gentle blur), **Sharpen**, **Emboss** and **Edge** (edges emphasised on top of the picture). They are applied before the colours are chosen.

The Apple IIgs has a richer menu called **Preprocessing**, with seven filters and a control for each. Choose a filter and its control appears beside it.

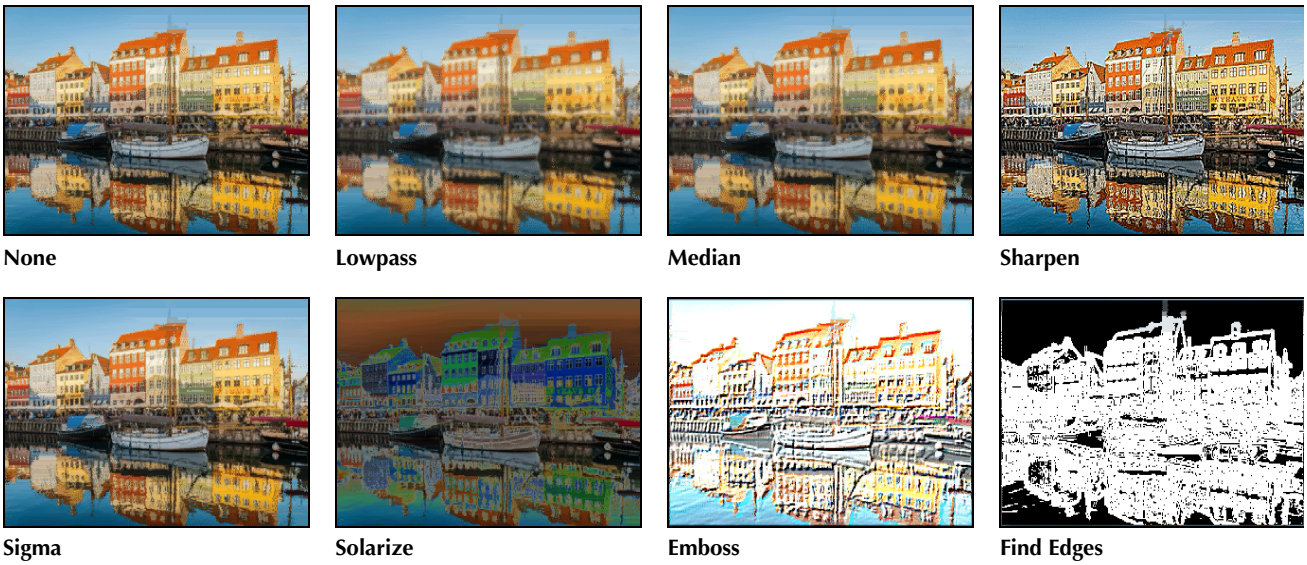


FIGURE 5-3
The Apple IIgs's preprocessing filters in 3,200-colour mode.

FILTER	CONTROL	WHAT IT DOES	GOOD FOR
Lowpass	none	A simple blur that averages neighbouring pixels.	Noise; softening harsh detail before conversion.
Median	Kernel Size: 3×3, 5×5, 7×7	Replaces each pixel with the median of its neighbours; removes noise while keeping edges.	Noisy or scanned pictures, JPEG artefacts.
Sharpen	Sharpen Amount: 0.2 to 2.5	Unsharp masking: increases local contrast along edges. 1.0 is standard; above 2 can create halos.	Soft or blurry pictures.
Sigma	Sigma Range: 5 to 50	An edge-preserving blur (a bilateral filter): smooths similar colours but keeps sharp boundaries.	Colour banding, smooth gradients, heavy compression.
Solarize	Threshold: 32 to 224	Inverts the tones above the threshold, like the darkroom effect.	Artistic effects.
Emboss	Emboss Depth: 0.3 to 2.0	A raised, three-dimensional relief effect.	Artistic effects, metallic looks.
Find Edges	Edge Sensitivity: 10 to 100	Sobel edge detection; the result looks like line art. Low values find many edges, high values only the strong ones.	Line-art extraction.
Edge (Filter menu)	none	Sobel edge detection blended with the original, rather than replacing it.	Making a flat picture look sharper.

5.6 Colour matching

Whenever BitPast has to decide which palette colour is closest to a colour in your picture, it needs a way to measure "closest". **Color Match** chooses the measure. The default, *Perceptive*, is right for most pictures.

METHOD	WHAT IT FAVOURS
Euclidean	Plain distance in RGB. Simple and fast.
Perceptive	Weighted for how people see colour. Recommended.
Luma	Matches brightness first.

METHOD	WHAT IT FAVOURS
Chroma	Matches colour and saturation first.
Hue	Keeps the hue. Best for saturated pictures and for fixed, saturated palettes.
Mahalanobis	A statistical distance (offered by some systems).



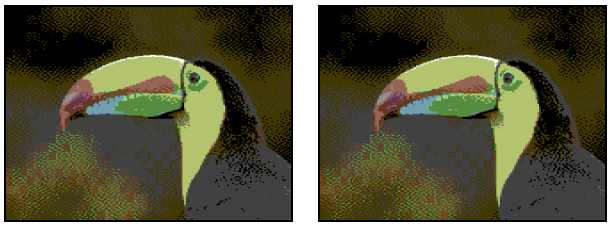
Euclidean **Perceptive** **Luma** **Chroma** **Hue**

FIGURE 5-4
Colour matching on the PC's fixed 16-colour EGA palette.

TIP With a fixed, strongly saturated palette — EGA, the BBC Micro, the MSX1 screen — Hue or Chroma often keeps colours that Perceptive turns into grey.

5.7 Pixel merge

In the modes where a pixel is twice as wide as it is high — C64 Multicolor, VIC-20 LowRes, Plus/4 Multicolor and CPC Mode 0 — two neighbouring dots of your picture become one machine pixel. **Pixel Merge** decides how: **Average** blends the two, for a smoother look; **Brightest** keeps the brighter one, which preserves highlights.



Average **Brightest**

FIGURE 5-5
Pixel Merge on the Commodore 64 in Multicolor mode.

5.8 Saturation and gamma

Saturation Changes the intensity of the colours before they are matched. 1 leaves them as they are. A small boost, 1.1 to 1.2, helps most pictures (the Apple IIgs starts at 1.10); beyond about 1.5 the colours become very strong and may clip.

Gamma Changes the brightness of the mid-tones, not black and white. Below 1 brightens them, above 1 darkens them; useful values lie between 0.8 and 1.2.

CHAPTER 6

6 The Systems

Fifteen machines, every mode and every setting

This chapter has one section for each of the fifteen systems, in the order of the system bar. Each section gives the system's facts, shows BitPast with the system selected, lists every mode with its picture, its file and the best use for it, and finishes with the settings that system offers.

Reading a section

The box in the margin of each section gives the essentials: the year of introduction, the video hardware, the colours and sizes, the native files and the keyboard shortcut. The pictures of the modes were all made from one photograph, with each mode's default settings, so you can compare them.

Screen shapes

The pictures are shown on a 4:3 screen, as the machines showed them. A picture that looks narrow or wide in the preview is shown here in its true proportions.

SYSTEM	INTRODUCED	VIDEO HARDWARE	SHORTCUT
Apple II	1977	NTSC artifact colour (b2d)	⇧⌘1
Apple IIgs	1986	Super Hi-Res	⇧⌘2
Amiga 500	1987	OCS/ECS chipset	⇧⌘3
Amiga 1200	1992	AGA chipset	⇧⌘4
Amstrad CPC	1984	Gate Array	⇧⌘5
Atari 800	1979	ANTIC and GTIA	⇧⌘6
Atari ST	1985	Shifter	⇧⌘7
BBC Micro	1981	6845 CRTC	⇧⌘8
Commodore 64	1982	VIC-II	⇧⌘9
MSX	1983	TMS9918 / V9938	⇧⌘0
IBM PC	1981	CGA, EGA, VGA	⇧⌘P
Commodore Plus/4	1984	TED	⇧⌘L
TRS-80 Color Computer	1980	MC6847 and GIME	⇧⌘T
VIC-20	1980	VIC	⇧⌘V
ZX Spectrum	1982	ULA	⇧⌘Z

6.1 Apple II

NTSC ARTIFACT COLOUR · CONVERTED WITH B2D

INTRODUCED
1977

VIDEO HARDWARE
NTSC artifact colour (b2d)

COLOURS
6 in Hi-Res, 16 in Double Hi-Res

NATIVE FILES
.A2FC .BIN .SLO .DLO .A2FM

SHORTCUT
⇧⌘1

The Apple II makes its colours in a way no other computer does: colour comes from where a dot lands on the NTSC television signal, so what you see depends on the pattern of the dots. BitPast converts Apple II pictures with *b2d* (BMP2DHR), Bill Buckels' converter, which is built into the app and understands this artifact colour.

Choose the screen format with **Target Format** and the size of the working bitmap with **Output Resolution**. The Apple II is the one system whose settings differ from all the others: b2d brings its own error-diffusion methods, ten colour tables and three tuning sliders.

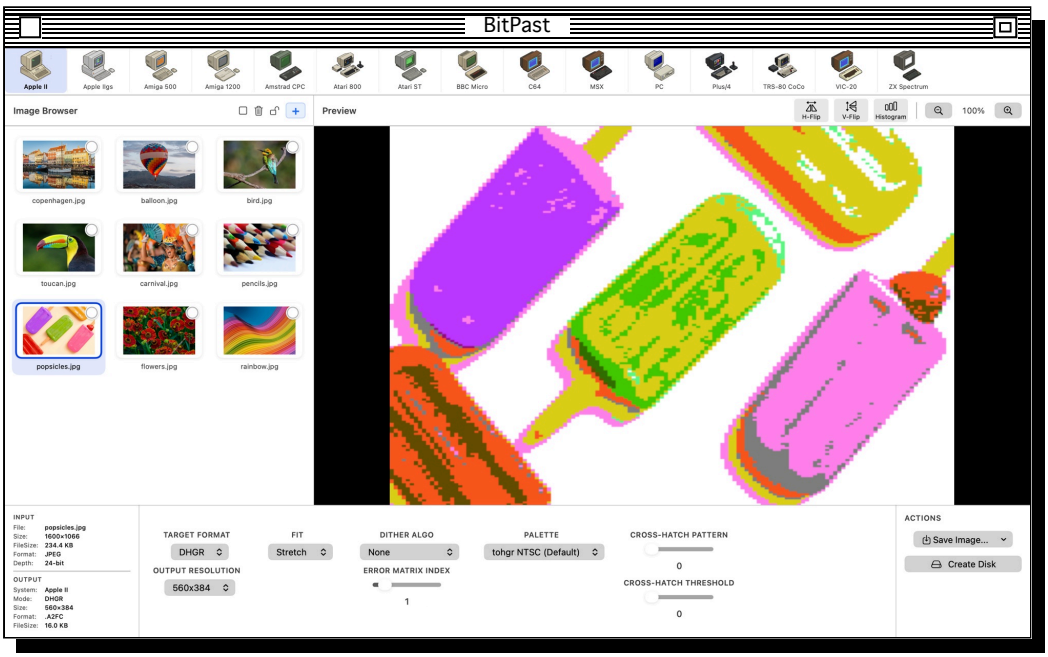
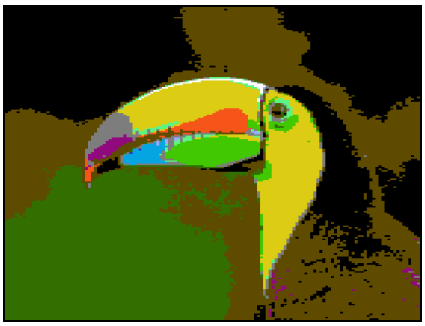


FIGURE 6-1
BitPast with the Apple II selected.

Modes

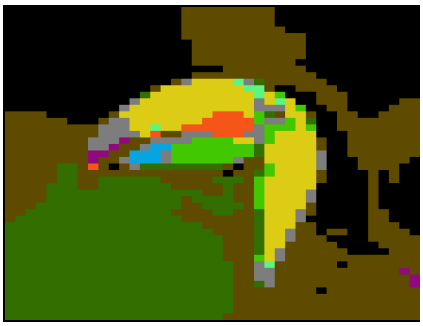
MODE	SCREEN	COLOURS	FILE	SIZE
DHGR (Double Hi-Res)	140×192 colour pixels (560 dots)	16	`.A2FC`	16,384 bytes
HGR (Hi-Res)	280×192	6	`.BIN`	8,192 bytes
LGR (Lo-Res)	40×48 blocks	16	`.SLO`	962 bytes
DLGR (Double Lo-Res)	80×48 blocks	16	`.DLO`	1,922 bytes
Mono (Monochrome)	560×192 or 280×192	2 (black and white)	`.A2FM`	16,384 bytes
… with 280×192	280×192	2	`.BIN`	8,192 bytes



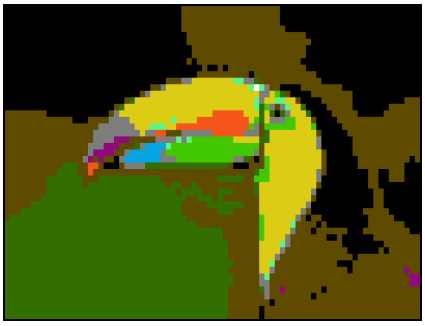
DHGR



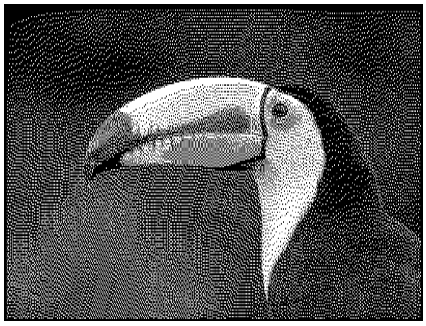
HGR



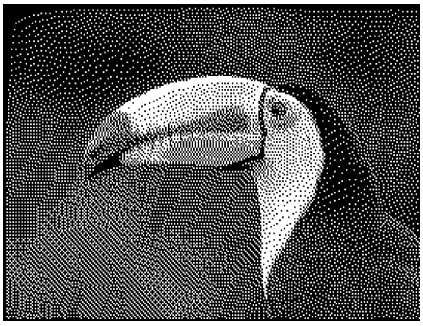
LGR



DLGR



Mono (560×192)



Mono (280×192)

DHGR (Double Hi-Res)

140×192 colour pixels (560 dots) · 16 · `.A2FC`

Double Hi-Res uses both banks of the Apple II's memory and shows 16 colours. Each colour pixel is four dots wide; the picture has 140 of them across, with the dots in between available for fine shading.

BEST FOR Photographs and artwork on a machine with 128 KB: an Apple IIe, IIc or IIgs.

- Floyd-Steinberg is recommended for photographs. Choose 560×384 as Output Resolution for the best result.

HGR (Hi-Res)

280×192 · 6 · `.BIN`

Hi-Res is the classic Apple II graphics mode: 280×192 dots and six colours (black, white, green, violet, orange and blue). The colours come from the horizontal position of each dot and from the dot patterns next to it.

BEST FOR Classic Apple II graphics, games, and artwork that needs an authentic period look.

- Error diffusion works well; Floyd-Steinberg is recommended for photographs.

LGR (Lo-Res)

40×48 blocks · 16 · `.SLO`

Lo-Res is the Apple II's block graphics: 40 by 48 coloured blocks, each one of 16 colours. The picture is tiny and chunky, and instantly recognisable.

BEST FOR Chunky, abstract pictures; the smallest Apple II files.

DLGR (Double Lo-Res)

80×48 blocks · 16 · `.DLO`

Double Lo-Res doubles the number of blocks across the screen to 80, using the second memory bank of a 128 KB machine.

BEST FOR A little more detail than Lo-Res, in the same blocky style.

Mono (Monochrome)

560×192 or 280×192 · 2 (black and white) · `.A2FM`

Black and white only. With a 560×384 working bitmap BitPast writes a double hi-res monochrome picture (560×192 dots, `.A2FM`); with 280×192 it writes an ordinary hi-res monochrome picture (`.BIN`).

BEST FOR Line art, text and pictures that should look like a green-screen or black-and-white monitor.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Target Format	DHGR, HGR, LGR, DLGR, Mono	DHGR	The graphics mode, described above.

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Output Resolution	140x192, 280x192, 320x200, 560x384, 640x400, 640x480	560x384	The size of the picture.
Dither Algo	None, Floyd-Steinberg, Jarvis, Stucki, Atkinson, Burkes, Sierra, Sierra Two, Sierra Lite, Buckels	None	How colours that the machine can't show are approximated (Chapter 5).
Palette	Kegs32 RGB, CiderPress RGB, AppleWin Old NTSC, AppleWin New NTSC, Wikipedia NTSC, tohgr NTSC (Default), Super Convert RGB, Jace NTSC, Cybernesto NTSC, tohgr NTSC HGR	tohgr NTSC (Default)	The colour table used to display and judge the Apple II's colours.
Cross-hatch Pattern	slider, 0 to 10	0	A b2d pattern setting; 0 turns it off.
Cross-hatch Threshold	slider, 0 to 40	0	A b2d threshold setting for the cross-hatch pattern.
Error Matrix Index	slider, 0 to 10	1	A b2d setting for the error-diffusion matrix.

The Apple II list is different from the others. **Dither Algo** offers b2d's nine error-diffusion methods (the choice of ordered dithers is not available), **Palette** chooses which table of NTSC and RGB colours is used to display and judge the colours — the default is *tohgr NTSC* — and the three sliders are passed straight to b2d: **Cross-hatch Pattern** (0 turns it off), **Cross-hatch Threshold** and **Error Matrix Index**. Leave them alone unless you want to experiment; the b2d documentation by Bill Buckels describes them in detail.

6.2 Apple IIgs

SUPER HI-RES · UP TO 3,200 COLOURS

INTRODUCED
1986

VIDEO HARDWARE
Super Hi-Res

COLOURS
16 per line; up to 3,200 in all

NATIVE FILES
.3200 .SHR

SHORTCUT
⇧⌘2

The Apple IIgs shows pictures on its Super Hi-Res screen: 320×200 pixels with 16 colours chosen from 4,096, or 640×200 pixels with 4. The hardware can change palette from line to line. A *scan control byte* (SCB) for each of the 200 lines picks one of 16 stored palettes — and the Brooks 3,200-colour format goes further and stores a full palette for every line, so a single picture can show 3,200 colours.

The IIgs converter is written natively in Swift. It is the only one with a palette editor (Chapter 7) and the only one with an extra choice of how palettes are shared between lines: **SCB Mode**.

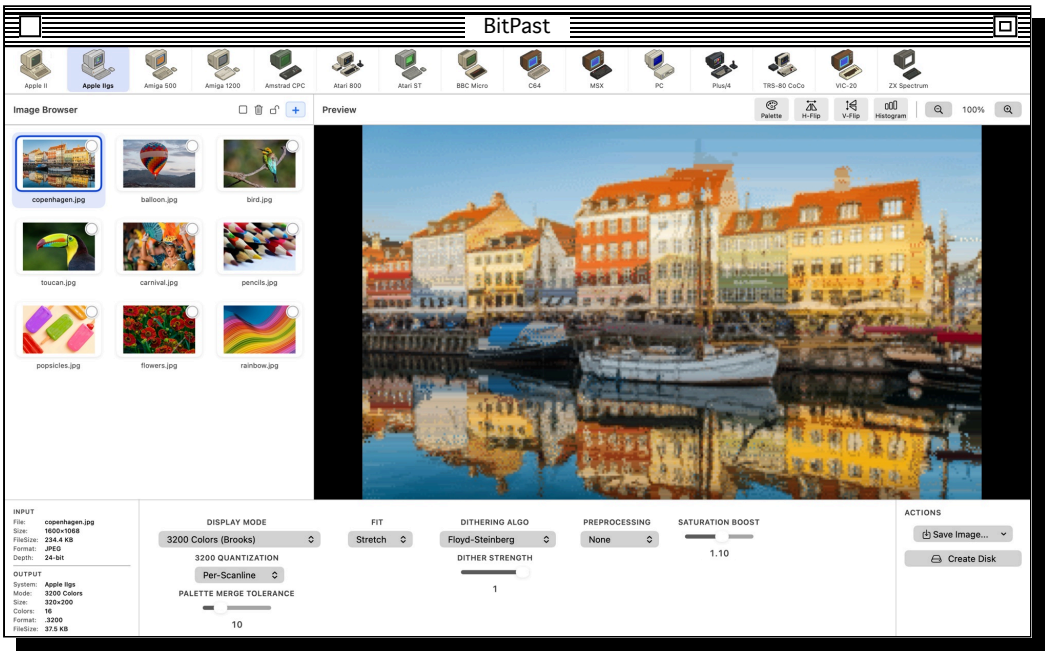
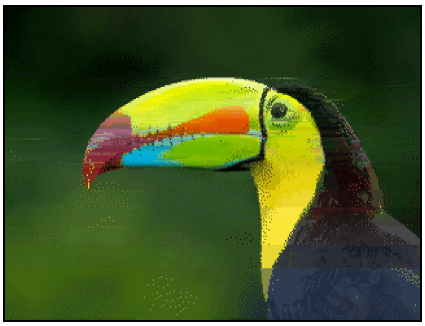


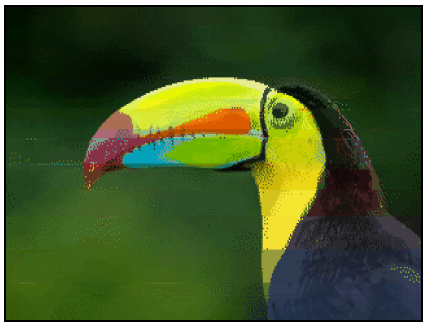
FIGURE 6-2
BitPast with the Apple IIgs selected.

Modes

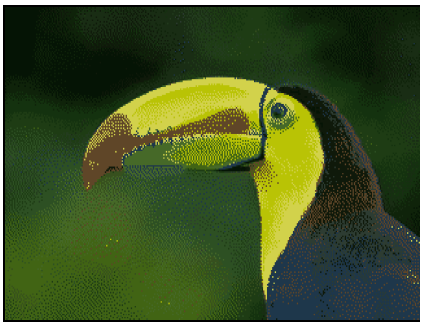
MODE	SCREEN	COLOURS	FILE	SIZE
3200 Colors (Brooks)	320×200	3,200 (16 per line)	.3200	38,400 bytes
256 Colors (16 Palettes)	320×200	256 (16 palettes of 16)	.SHR	32,768 bytes
320x200 (16 Colors)	320×200	16	.SHR	32,768 bytes
640x200 (4 Colors)	640×200	4	.SHR	32,768 bytes
640x200 Enhanced (16 Colors)	640×200	16 (8 effective)	.SHR	32,768 bytes
640x200 Desktop (16 Colors)	640×200	GS/OS Finder colours	.SHR	32,768 bytes



3200 Colors



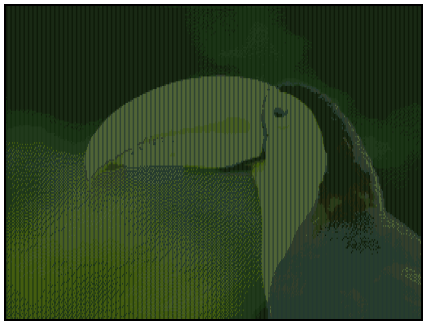
256 Colors (Per-Line)



320×200, 16 colors



640×200, 4 colors



640×200 Enhanced



640×200 Desktop

3200 Colors (Brooks)

320×200 · 3,200 (16 per line) · `.3200`

True 3200-color mode using the Brooks format. Each of the 200 scanlines has its own independent 16-color palette, allowing up to 3,200 unique colors on screen.

BEST FOR Photographic images, gradients, images with many colors

- **Palette Merge Tolerance:** Controls palette reuse between similar scanlines. Higher values reduce banding but may lose color accuracy. Set to 0 for maximum color fidelity, 10-20 for balanced results.
- **Dithering:** All dithering algorithms work well. Floyd-Steinberg is recommended for photos.
- **Saturation Boost:** Increase to compensate for the IIgs's somewhat muted palette.

---

256 Colors (16 Palettes)

320×200 · 256 (16 palettes of 16) · `.SHR`

Standard SHR format with 16 palette slots. Scanlines are grouped and share palettes. Maximum 256 unique colors (16 palettes × 16 colors).

BEST FOR Compatibility with standard SHR viewers, when file size matters

- **SCB Mode:** Which of the 16 palettes each scanline uses. Banded: 16 even bands of lines. Palette Reuse: the older method, which reuses the previous line's palette when it fits well enough; on photos it usually looks worse than Banded. Per-Line: any line can use any palette, so similar lines share one even when they're far apart; usually the best quality.
- **Palette Merge Tolerance:** Only visible with 'Palette Reuse'. Controls how similar scanlines must be to share a palette.

320x200 (16 Colors)

320×200 · 16 · `.SHR`

Standard Super Hi-Res mode with a single 16-color palette for the entire image. Classic IIgs graphics mode.

BEST FOR Simple graphics, logos, images with limited color range

- **Dithering:** Essential for this mode. Atkinson gives a classic Mac look, Floyd-Steinberg for smoother results.

640x200 (4 Colors)

640×200 · 4 · `.SHR`

High-resolution mode with 640 horizontal pixels but only 4 colors. Good for text and line art.

BEST FOR Screenshots, text-heavy images, technical drawings

- **Dithering:** Ordered (Bayer) dithering works well to avoid artifacts.

640x200 Enhanced (16 Colors)

640×200 · 16 (8 effective) · .SHR

Enhanced 640 mode using column-aware dithering. Even columns use one 4-color sub-palette, odd columns use another, for 8 effective colors.

BEST FOR Higher resolution images that need more color variety

- **Dithering:** Floyd-Steinberg recommended for best results.

640x200 Desktop (16 Colors)

640×200 · GS/OS Finder colours · .SHR

Uses the GS/OS Finder palette with column-aware dithering. Even columns: Black, Blue, Yellow, White. Odd columns: Black, Red, Green, White.

BEST FOR Images that should match the GS/OS desktop aesthetic

- **Dithering:** All algorithms work, but results will be limited to the fixed Finder palette.

SCB Mode

In the 256-colour mode the hardware has 16 palettes, and every one of the 200 lines uses one of them. **SCB Mode** decides which. Its three choices give three different pictures from the same photograph.

SCB MODE	HOW LINES ARE GIVEN PALETTES
Banded	The screen is divided into 16 even bands of lines, and every band has a palette of its own. Fast and predictable. This is the default.
Palette Reuse	The older method. A line keeps the previous line's palette when that fits well enough (see Palette Merge Tolerance), and starts a new palette when it does not. On photographs it usually looks worse than Banded.
Per-Line	Any line can use any of the 16 palettes. BitPast starts from 16 bands, gives each line the palette that fits it best, rebuilds the palettes from the lines assigned to them, and repeats until nothing changes. Similar lines share a palette even when they are far apart, so this usually gives the best quality.



Banded
16 even bands.



Palette Reuse
Reuses the previous line's palette.



Per-Line
Any line, any palette.

FIGURE 6-3
The three SCB modes on the same picture. Look at the sky.

TIP To fine-tune individual palettes after a conversion, use the palette editor (Chapter 7).

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Display Mode	3200 Colors (Brooks), 256 Colors (16 Palettes), 320x200 (16 Colors), 640x200 (4 Colors), 640x200 Enhanced (16 Colors), 640x200 Desktop (16 Colors)	3200 Colors (Brooks)	The graphics mode, described above.
Dithering Algo	None, Floyd-Steinberg, Atkinson, Jarvis-Judice-Ninke, Stucki, Burkes, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue 8x8, Blue 16x16	Floyd-Steinberg	How colours that the machine can't show are approximated (Chapter 5).

SETTING	CHOICES	DEFAULT	WHAT IT DOES
3200 Quantization	Per-Scanline, Palette Reuse	Per-Scanline	How palettes are chosen in 3,200-colour mode.
SCB Mode	Banded, Palette Reuse, Per-Line	Banded	Which of the 16 palettes each line uses in 256-colour mode.
Dither Strength	slider, 0 to 1	1	How strongly the dither is applied; 0 is none, 1 is full strength.
Palette Merge Tolerance	slider, 0 to 50	10	How alike two lines must be to share a palette. Shown with Palette Reuse.
Saturation Boost	slider, 0 to 2	1.10	Colour intensity; 1 leaves it unchanged.
Preprocessing	None, Lowpass, Median, Sharpen, Sigma, Solarize, Emboss, Find Edges	None	A filter applied before converting (Chapter 5).
Kernel Size	3x3, 5x5, 7x7	3x3	Size of the area used by the Median filter.
Sharpen Amount	slider, 0.2 to 2.5	1	Strength of the Sharpen filter.
Sigma Range	slider, 5 to 50	20	How different colours may be and still be smoothed together by the Sigma filter.
Threshold	slider, 32 to 224	128	Brightness above which the Solarize filter inverts tones.
Emboss Depth	slider, 0.3 to 2	1	Depth of the Emboss effect.
Edge Sensitivity	slider, 10 to 100	50	How strong an edge must be for Find Edges to draw it.

3200 Quantization applies to the 3,200-colour mode: *Per-Scanline* builds a new palette for every line, *Palette Reuse* lets a line keep the previous line's palette when it fits well enough (see **Palette Merge Tolerance**), which reduces banding where neighbouring lines are alike. **SCB Mode** applies to the 256-colour mode and is explained below. **Preprocessing** offers the seven filters described in Chapter 5.

6.3 Amiga 500

OCS/ECS · 32 COLOURS · HAM6

INTRODUCED
1987

VIDEO HARDWARE
OCS/ECS chipset

COLOURS
32 of 4,096; 4,096 in HAM6

NATIVE FILES
.iff

SHORTCUT
⇧⌘3

The Amiga 500's Original Chip Set shows up to 32 colours at a time, chosen from a palette of 4,096 (16 levels each of red, green and blue). Its famous trick is *HAM6*, "hold and modify": each pixel either takes a colour from a small palette or changes one colour component of the pixel to its left, which puts all 4,096 colours on one screen.

BitPast saves standard IFF/ILBM pictures, the format Amiga painting programs and emulators load. The picture is 320 pixels wide and either 256 lines high (the PAL screen) or 512 lines (interlaced).

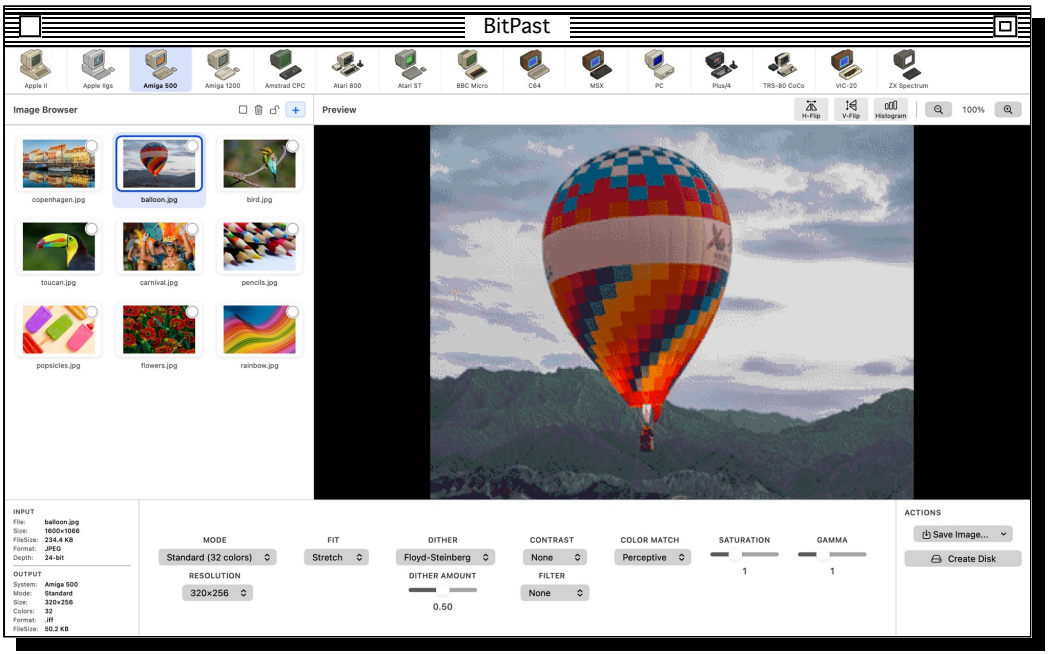


FIGURE 6-4
BitPast with the Amiga 500 selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
Standard (32 colors)	320×256 or 320×512	32 of 4,096	`.iff`	varies
HAM6 (4096 colors)	320×256 or 320×512	4,096	`.iff`	varies



Standard (32 colors)



HAM6 (4096 colors)

Standard (32 colors)

320×256 or 320×512 · 32 of 4,096 · `.iff`

Standard Amiga OCS mode with 32 colors selected from the 4096-color hardware palette. Uses 5 bitplanes for 32 simultaneous colors.

BEST FOR General graphics, games, images with limited color range

- **Resolution:** 320×256 (PAL standard) or 320×512 (interlaced)
- **Dithering:** Bayer 4×4 is default. Error diffusion also works well.
- **Color Matching:** Perceptive recommended. Automatically selects optimal 32 colors.

HAM6 (4096 colors)

320×256 or 320×512 · 4,096 · `.iff`

Hold-And-Modify mode allows up to 4096 colors on screen. Each pixel can either use a palette color or modify one RGB channel from the previous pixel. Uses 6 bitplanes.

BEST FOR Photographs, gradients, images requiring many colors

- **Resolution:** 320×256 (PAL standard) or 320×512 (interlaced)
- **Dithering:** Usually not needed due to high color count. Subtle dithering can help with gradients.

NOTE HAM changes only one colour component from one pixel to the next, so a sharp edge between two very different colours can show a coloured fringe. This is typical of the mode, and is why HAM suits smooth photographs better than line art.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Mode	Standard (32 colors), HAM6 (4096 colors)	Standard (32 colors)	The graphics mode, described above.
Resolution	320×256, 320×512	320×256	The size of the picture.
Dither	None, Floyd-Steinberg, Atkinson, Bayer 2x2, Bayer 4x4, Bayer 8x8	Floyd-Steinberg	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0.5 to 2	1	Colour intensity; 1 leaves it unchanged.

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.4 Amiga 1200

AGA · 256 COLOURS · HAM8

INTRODUCED
1992

VIDEO HARDWARE
AGA chipset

COLOURS
256 of 16.7 million; 262,144 in HAM8

NATIVE FILES
.iff

SHORTCUT
⇧⌘4

The Amiga 1200's Advanced Graphics Architecture lifts the palette to 24 bits (16.7 million colours) and the number of colours on screen to 256. HAM8, the improved hold-and-modify mode, can show 262,144 colours at once with six bits for each colour component.

Choose between three screen sizes with **Resolution**: 320×256, 320×512 (interlaced) and 640×512 (high-resolution interlaced).

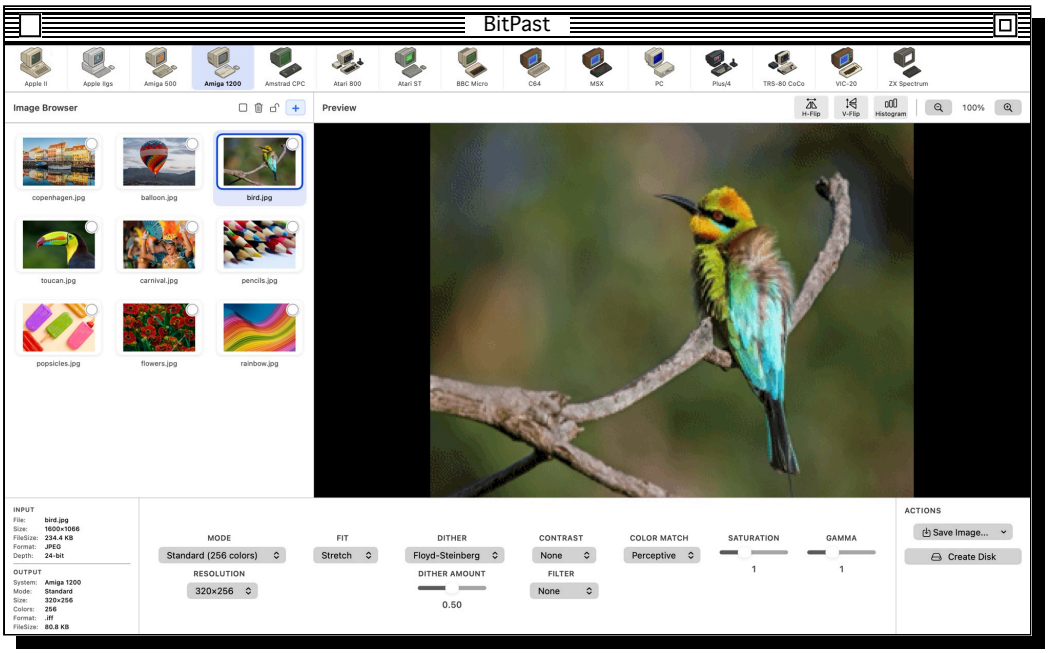


FIGURE 6-5
BitPast with the Amiga 1200 selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
Standard (256 colors)	320×256, 320×512 or 640×512	256 of 16.7 million	`.iff`	varies
HAM8 (262144 colors)	320×256, 320×512 or 640×512	262,144	`.iff`	varies



Standard (256 colors)



HAM8 (262144 colors)

Standard (256 colors)

320×256, 320×512 or 640×512 · 256 of 16.7 million · `.iff`

AGA mode with 256 colors selected from the full 24-bit palette. Uses 8 bitplanes for 256 simultaneous colors.

BEST FOR General graphics, games, images with moderate color range

- **Resolution:** 320×256, 320×512 (interlaced), or 640×512 (hi-res interlaced)
- **Dithering:** Bayer 4×4 is default. With 256 colors, less dithering is usually needed.
- **Color Matching:** Perceptive recommended. Automatically selects optimal 256 colors.

HAM8 (262144 colors)

320×256, 320×512 or 640×512 · 262,144 · `.iff`

AGA Hold-And-Modify mode allows up to 262,144 colors on screen. Each pixel can either use a palette color or modify one RGB channel (6 bits per channel) from the previous pixel. Uses 8 bitplanes.

BEST FOR Photographs, photorealistic images, gradients

- **Resolution:** 320×256, 320×512 (interlaced), or 640×512 (hi-res interlaced)
- **Dithering:** Rarely needed. HAM8 provides excellent color reproduction.



320×256



320×512 (interlaced)



640×512 (hi-res interlaced)

FIGURE 6-6

The three resolutions. More pixels give more detail, but bigger files.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Mode	Standard (256 colors), HAM8 (262144 colors)	Standard (256 colors)	The graphics mode, described above.
Resolution	320×256, 320×512, 640×512	320×256	The size of the picture.
Dither	None, Floyd-Steinberg, Atkinson, Bayer 2x2, Bayer 4x4, Bayer 8x8	Floyd-Steinberg	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0.5 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.5 Amstrad CPC

GATE ARRAY · 27-COLOUR PALETTE

INTRODUCED
1984

VIDEO HARDWARE
Gate Array

COLOURS
4 or 16 of 27

NATIVE FILES
.scr

SHORTCUT
⇧⌘5

The Amstrad CPC has a hardware palette of 27 colours (three levels each of red, green and blue). In Mode 1, four of them are on screen at once at 320×200. In Mode 0, sixteen are on screen at 160×200, with pixels twice as wide. BitPast chooses the best four or sixteen colours for each picture.

Files are written with an AMSDOS header, which is what lets them load from a CPC disk image.

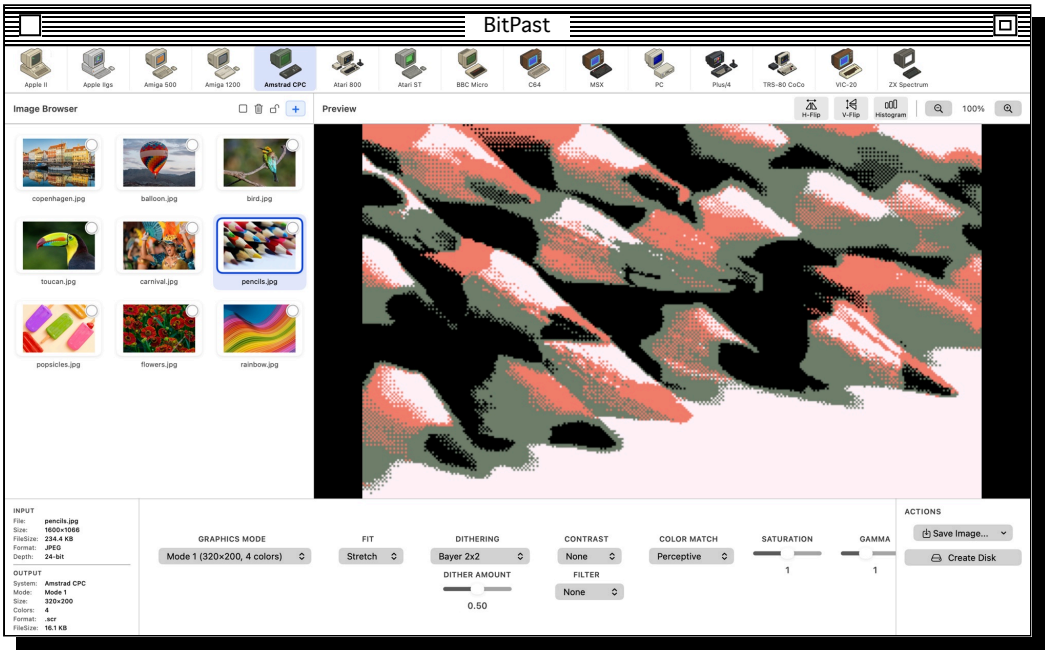
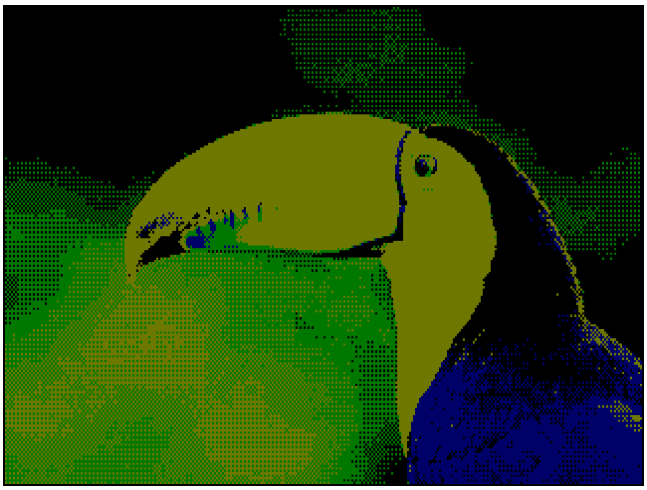


FIGURE 6-7
BitPast with the Amstrad CPC selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
Mode 1 (320×200, 4 colors)	320×200	4 of 27	.scr	16,512 bytes
Mode 0 (160×200, 16 colors)	160×200	16 of 27	.scr	16,512 bytes



Mode 1



Mode 0

Mode 1 (320×200, 4 colors)

320×200 · 4 of 27 · `.scr`

Standard CPC mode with 320×200 resolution and 4 colors selected from the 27-color hardware palette. Good balance of resolution and color.

BEST FOR General purpose graphics, games, detailed images

- **Dithering:** Error diffusion recommended. Floyd-Steinberg or Atkinson work well.
- **Color Matching:** The converter automatically selects optimal 4 colors from 27.

Mode 0 (160×200, 16 colors)

160×200 · 16 of 27 · `.scr`

Low-resolution CPC mode with double-wide pixels. Uses all 16 palette entries, selected from 27 hardware colors. Most colorful CPC mode.

BEST FOR Colorful images, photos, artwork needing many colors

- **Pixel Merge:** Average: Blends adjacent pixels. Brightest: Picks brighter pixel for highlights.
- **Dithering:** Ordered dithering often gives better results than error diffusion in this mode.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Graphics Mode	Mode 1 (320×200, 4 colors), Mode 0 (160×200, 16 colors)	Mode 1 (320×200, 4 colors)	The graphics mode, described above.
Dithering	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue 8x8, Blue 16x16	Bayer 2x2	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Pixel Merge	Average, Brightest	Average	How the two halves of a double-wide pixel are combined: Average blends them, Brightest keeps the brighter one.
Color Match	Euclidean, Perceptive, Luma, Chroma, Mahalanobis	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.6 Atari 800

ANTIC/GTIA · 128-COLOUR PALETTE

INTRODUCED
1979

VIDEO HARDWARE
ANTIC and GTIA

COLOURS
2 to 16 of 128

NATIVE FILES
.gr8 .gr15 .gr9 .gr10 .gr11

SHORTCUT
⇧⌘6

The Atari 800's ANTIC and GTIA chips have a palette of 128 colours: 16 hues, each in 8 levels of brightness. The five modes BitPast offers have very different characters. Graphics 8 is a sharp two-colour mode (both colours are shades of one hue, as on the real hardware). Graphics 15 gives four colours in double-wide pixels. The three GTIA modes — 9, 10 and 11 — trade resolution (only 80 pixels across) for 16 shades, nine colours or 16 hues.

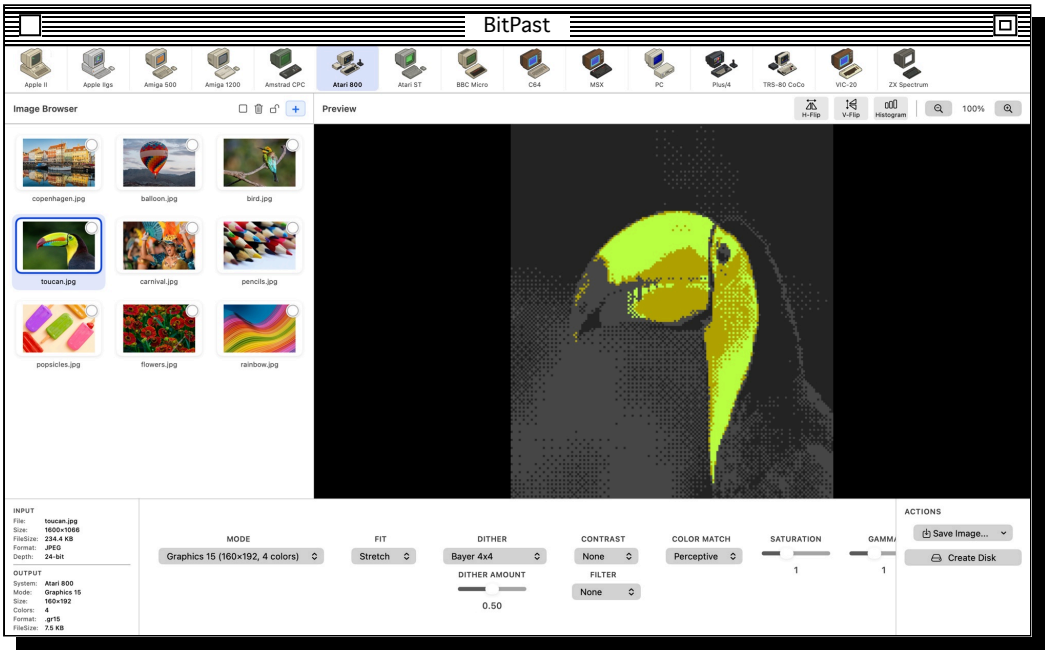
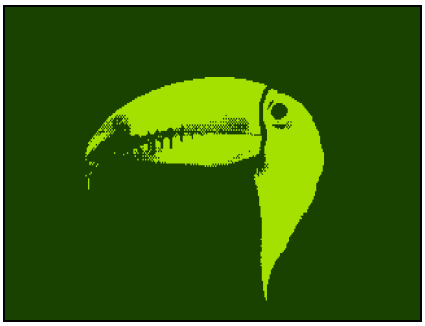


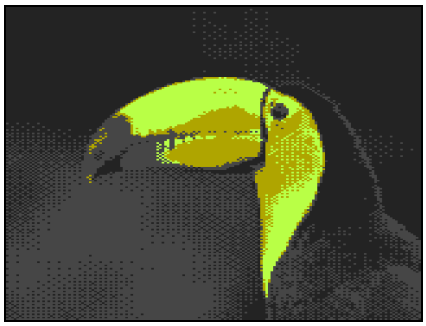
FIGURE 6-8
BitPast with the Atari 800 selected.

Modes

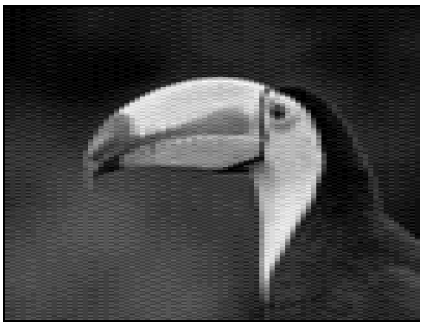
MODE	SCREEN	COLOURS	FILE	SIZE
Graphics 8 (320×192, 2 colors)	320×192	2	.gr8	7,682 bytes
Graphics 15 (160×192, 4 colors)	160×192	4 of 128	.gr15	7,684 bytes
Graphics 9 (80×192, 16 shades)	80×192	16 shades	.gr9	7,696 bytes
Graphics 10 (80×192, 9 colors)	80×192	9 of 128	.gr10	7,689 bytes
Graphics 11 (80×192, 16 hues)	80×192	16 hues	.gr11	7,696 bytes



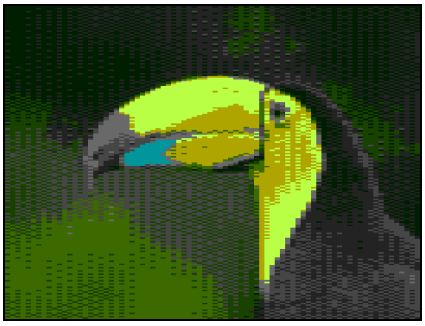
Graphics 8



Graphics 15



Graphics 9



Graphics 10



Graphics 11

Graphics 8 (320×192, 2 colors)

320×192 · 2 · `.gr8`

High-resolution monochrome mode. 320×192 pixels with 2 colors (background + foreground). Best for line art and high-contrast images.

BEST FOR Line art, text, high contrast images

- **Dithering:** Bayer dithering creates good patterns in monochrome.
- **Contrast:** HE or CLAHE can improve detail in photos.

Graphics 15 (160×192, 4 colors)

160×192 · 4 of 128 · `.gr15`

Medium-resolution mode with 160×192 pixels and 4 colors from the 128-color palette. Double-wide pixels. Most versatile mode for general images.

BEST FOR General graphics, photos, colorful images

- **Dithering:** Bayer 4×4 is default. Floyd-Steinberg also works well.
- **Color Matching:** Perceptive recommended for photos.

Graphics 9 (80×192, 16 shades)

80×192 · 16 shades · `.gr9`

GTIA mode with 80×192 pixels and 16 shades of grayscale. Quad-wide pixels. Excellent for photographic images.

BEST FOR Photos, portraits, grayscale images

- **Dithering:** Minimal dithering often best. Try Floyd-Steinberg for smoother gradients.
- **Contrast:** CLAHE can enhance detail in low-contrast photos.

Graphics 10 (80×192, 9 colors)

80×192 · 9 of 128 · `.gr10`

GTIA color mode with 80×192 pixels and 9 colors selected from the 128-color palette. Quad-wide pixels. Good for colorful images with limited detail needs.

BEST FOR Colorful low-res images, artistic graphics

- **Dithering:** Bayer dithering recommended. Error diffusion may produce visible artifacts.
- **Color Matching:** Perceptive recommended for natural color selection from the 128-color palette.

Graphics 11 (80×192, 16 hues)

80×192 · 16 hues · `.gr11`

GTIA hue mode with 80×192 pixels and 16 different hues at one luminance level. Quad-wide pixels. All colors have the same brightness.

BEST FOR Images with many different colors but uniform brightness, artistic effects

- **Dithering:** Bayer dithering works well. Hue-based dithering preserves color variety.
- **Note:** All 16 colors share the same luminance, so contrast comes from hue differences only.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Mode	Graphics 8 (320×192, 2 colors), Graphics 15 (160×192, 4 colors), Graphics 9 (80×192, 16 shades), Graphics 10 (80×192, 9 colors), Graphics 11 (80×192, 16 hues)	Graphics 15 (160×192, 4 colors)	The graphics mode, described above.
Dither	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue Noise 8x8, Blue Noise 16x16	Bayer 4x4	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma, Hue	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0.5 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.7 Atari ST

SHIFTER · 512-COLOUR PALETTE

INTRODUCED
1985

VIDEO HARDWARE
Shifter

COLOURS
16, 4 or 2 of 512

NATIVE FILES
.pi1 .pi2 .pi3

SHORTCUT
⇧⌘7

The Atari ST's palette has 512 colours (eight levels each of red, green and blue). Low resolution shows 16 of them at 320×200, medium resolution shows 4 at 640×200, and high resolution is black and white at 640×400 on the ST's monochrome monitor. BitPast saves DEGAS Elite pictures (.PI1, .PI2, .PI3).

Medium resolution is a good place to see what **Aspect** does: its pixels are very tall, so a picture placed with Fit set to Contain looks quite different with Aspect set to Correct or Ignore (Chapter 4).

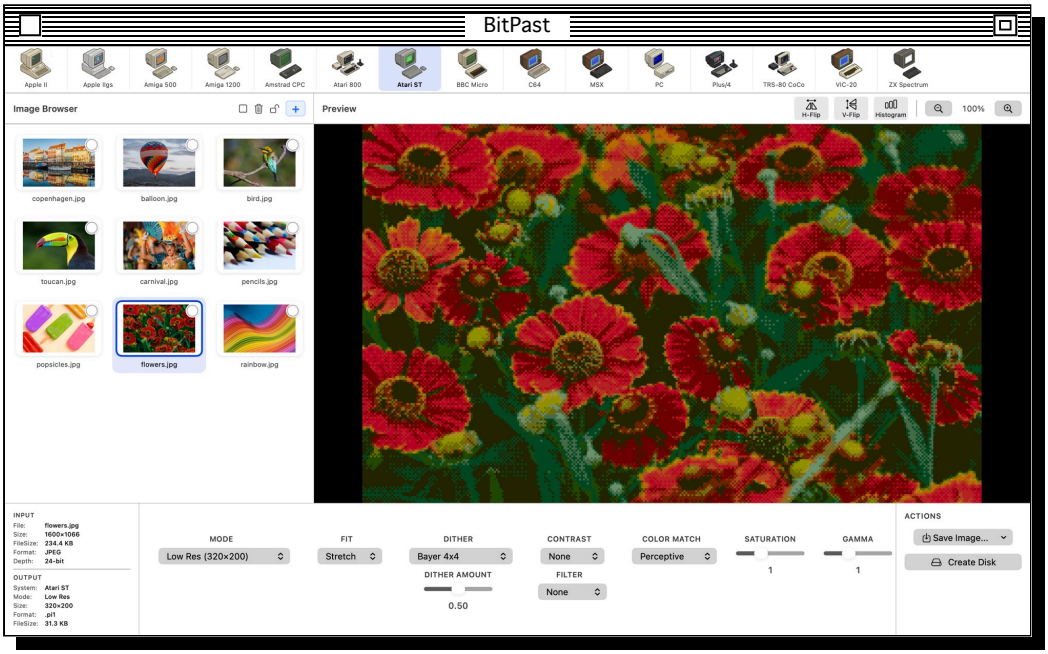
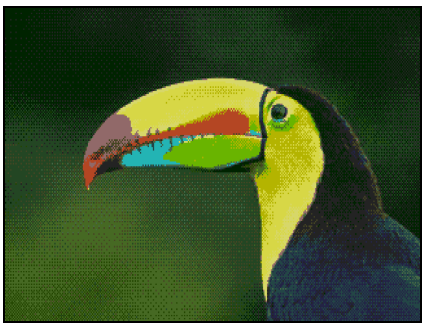


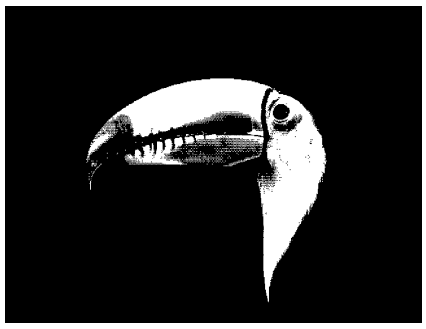
FIGURE 6-9
BitPast with the Atari ST selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
Low Res (320×200)	320×200	16 of 512	`.pi1`	32,034 bytes
Medium Res (640×200)	640×200	4 of 512	`.pi2`	32,034 bytes
High Res (640×400)	640×400	2 (black and white)	`.pi3`	32,034 bytes


Low Res


Medium Res


High Res

Low Res (320×200)

320×200 · 16 of 512 · `.pi1`

Atari ST low-resolution mode with 320×200 pixels and 16 colors selected from the 512-color hardware palette. Uses 4 bitplanes. Standard mode for colorful graphics.

BEST FOR General graphics, photos, colorful images

- **Dithering:** Bayer 4×4 is default. Error diffusion (Floyd-Steinberg, Atkinson) also works well.
- **Color Matching:** Perceptive recommended. The converter automatically selects optimal 16 colors from 512.

Medium Res (640×200)

640×200 · 4 of 512 · `.pi2`

Atari ST medium-resolution mode with 640×200 pixels and 4 colors selected from the 512-color hardware palette. Uses 2 bitplanes. Good balance of resolution and color.

BEST FOR Text, line art, graphics needing higher horizontal resolution

- **Dithering:** Ordered dithering (Bayer) often works well with limited 4-color palette.
- **Color Matching:** Perceptive recommended. Optimal 4 colors are automatically selected.

High Res (640×400)

640×400 · 2 (black and white) · `.pi3`

Atari ST high-resolution mode with 640×400 pixels in monochrome (2 colors: black and white). Uses 1 bitplane. Requires a dedicated high-res monitor on original hardware.

BEST FOR Text, CAD, desktop publishing, high-detail line art

- **Dithering:** Bayer dithering creates good halftone patterns. Floyd-Steinberg for photographs.
- **Contrast:** CLAHE or HE can help improve detail in photos before conversion.

Chapter 4 shows how Aspect changes a picture in medium resolution (Figure 4-2).

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Mode	Low Res (320×200), Medium Res (640×200), High Res (640×400)	Low Res (320×200)	The graphics mode, described above.
Dither	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue Noise 8x8, Blue Noise 16x16	Bayer 4x4	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma, Mahalanobis	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0.5 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.8 BBC Micro

6845 CRTC · 8 COLOURS

INTRODUCED
1981

VIDEO HARDWARE
6845 CRTC

COLOURS
2, 4 or 8

NATIVE FILES
.bbc

SHORTCUT
⇧⌘8

The BBC Micro has eight fixed colours — black, red, green, yellow, blue, magenta, cyan and white. How many are on screen at once, and at what resolution, depends on the mode. Mode 0 gives two colours at 640×256, Mode 1 four at 320×256, Mode 2 all eight at 160×256, and Modes 4 and 5 are the half-memory versions of Modes 1 and 2. BitPast writes the raw screen memory (`.bbc`).

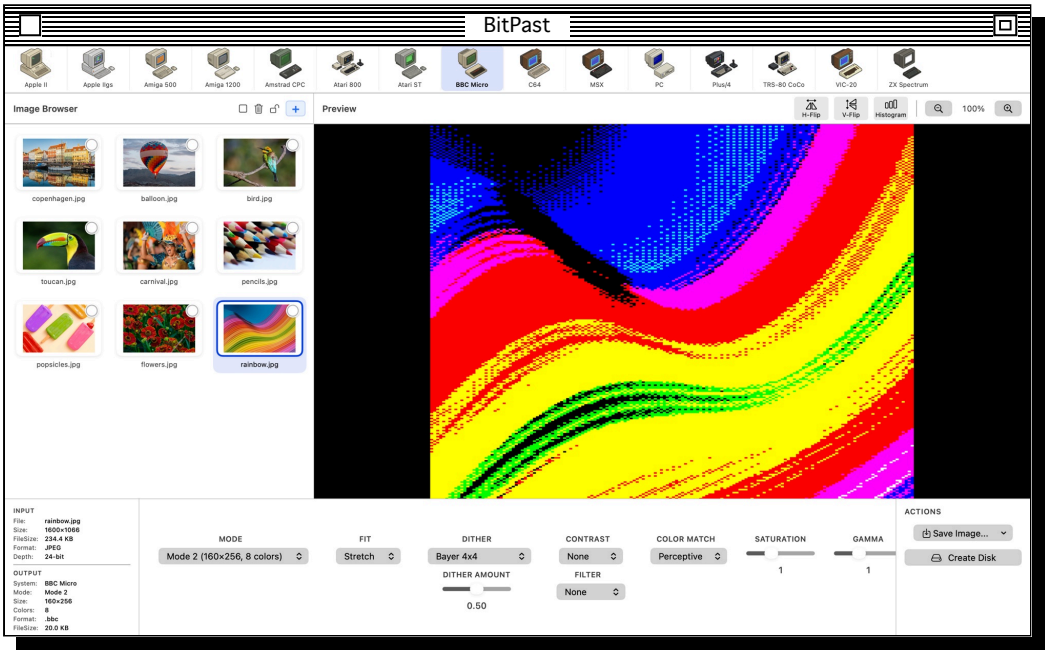
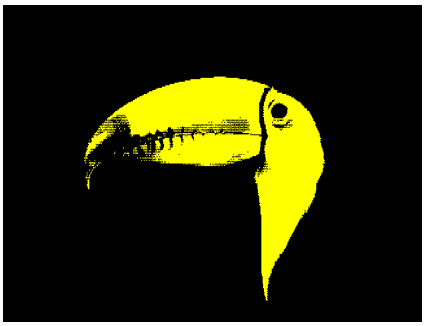


FIGURE 6-10
BitPast with the BBC Micro selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
Mode 0 (640×256, 2 colors)	640×256	2 of 8	.bbc	20,483 bytes
Mode 1 (320×256, 4 colors)	320×256	4 of 8	.bbc	20,485 bytes
Mode 2 (160×256, 8 colors)	160×256	8	.bbc	20,489 bytes
Mode 4 (320×256, 2 colors)	320×256	2 of 8	.bbc	10,243 bytes
Mode 5 (160×256, 4 colors)	160×256	4 of 8	.bbc	10,245 bytes



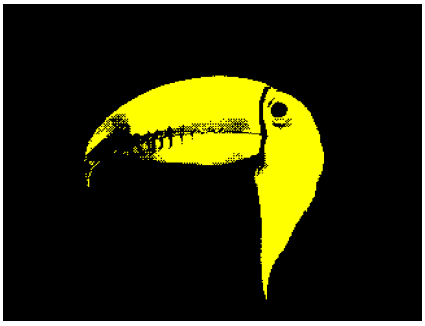
Mode 0



Mode 1



Mode 2



Mode 4



Mode 5

Mode 0 (640×256, 2 colors)

640×256 · 2 of 8 · `.bbc`

High-resolution monochrome mode with 640×256 pixels. 2 colors selected from the 8-color palette. 80 bytes per line.

BEST FOR Text, line art, high-resolution graphics

- **Dithering:** Bayer dithering creates good patterns in 2-color mode.
- **Contrast:** HE or CLAHE can improve detail in photos.

Mode 1 (320×256, 4 colors)

320×256 · 4 of 8 · `.bbc`

Medium-resolution mode with 320×256 pixels. 4 colors selected from the 8-color palette. Good balance of resolution and color.

BEST FOR General graphics, games, colorful images

- **Dithering:** Floyd-Steinberg or Bayer 4×4 recommended.
- **Color Matching:** Perceptive recommended for natural color selection.

Mode 2 (160×256, 8 colors)

160×256 · 8 · `.bbc`

Low-resolution mode with 160×256 pixels and all 8 colors available. Quad-wide pixels. Most colorful BBC Micro mode.

BEST FOR Colorful images, photos, artwork needing full palette

- **Dithering:** Often not needed with 8 colors. Light dithering can help gradients.
- **Color Matching:** Hue matching preserves colors well with this saturated palette.

Mode 4 (320×256, 2 colors)

320×256 · 2 of 8 · `.bbc`

Medium-resolution monochrome mode. Same resolution as Mode 1 but with only 2 colors, using half the memory.

BEST FOR Line art, text, when memory is limited

- **Dithering:** Bayer dithering recommended for photos.

Mode 5 (160×256, 4 colors)

160×256 · 4 of 8 · `.bbc`

Low-resolution mode with 4 colors. Same resolution as Mode 2 but with fewer colors, using half the memory.

BEST FOR Colorful graphics when memory is limited

- **Dithering:** Floyd-Steinberg recommended for photos.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Mode	Mode 0 (640×256, 2 colors), Mode 1 (320×256, 4 colors), Mode 2 (160×256, 8 colors), Mode 4 (320×256, 2 colors), Mode 5 (160×256, 4 colors)	Mode 2 (160×256, 8 colors)	The graphics mode, described above.
Dither	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue Noise 8x8, Blue Noise 16x16	Bayer 4x4	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma, Hue	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0.5 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.9 Commodore 64

VIC-II · 16 COLOURS

INTRODUCED
1982

VIDEO HARDWARE
VIC-II

COLOURS
16

NATIVE FILES
.art .kla .prg

SHORTCUT
⇧⌘9

Commodore 64 pictures follow the rules of the VIC-II chip. In HiRes mode every 8×8 cell of the picture may use two colours. In Multicolor mode every 4×8 cell may use four: three chosen for that cell and a background colour shared by the whole picture. The palette has 16 colours (the well-known Pepto palette).

BitPast saves Art Studio (`.art`) for HiRes, Koala Painter (`.kla`) for Multicolor and, for PETSCII, a program (`.prg`) that shows the picture when you run it.

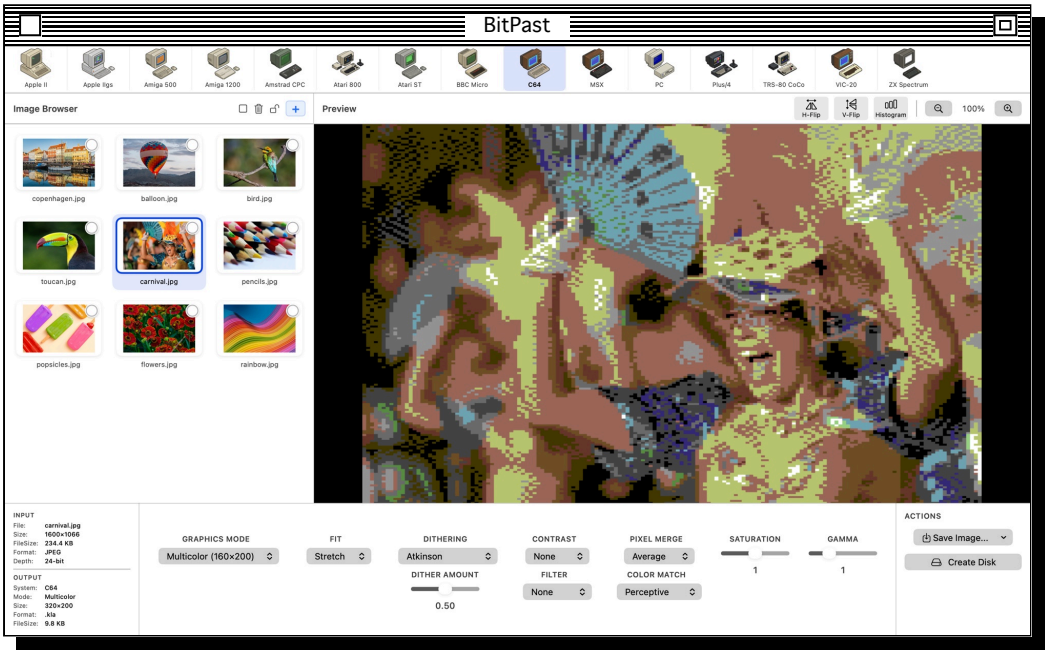
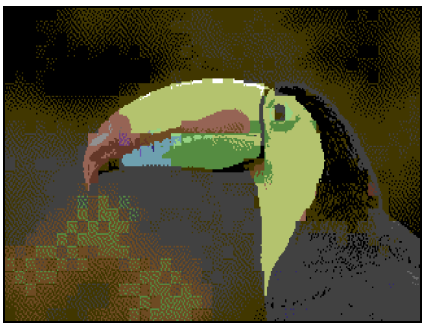
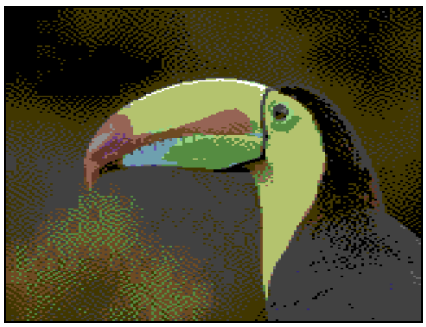


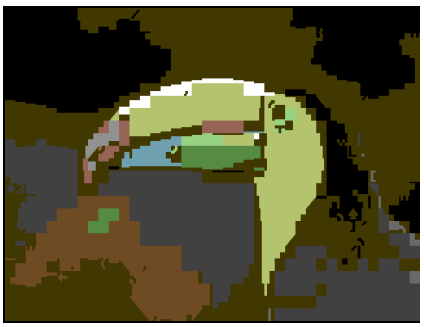
FIGURE 6-11
BitPast with the Commodore 64 selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
HiRes (320×200)	320×200	2 per 8×8 cell	`.art`	9,002 bytes
Multicolor (160×200)	160×200	4 per 4×8 cell	`.kla`	10,003 bytes
PETSCII (40×25)	40×25 characters	16 (2 per cell)	`.prg`	varies


HiRes


Multicolor


PETSCII

HiRes (320×200)

320×200 · 2 per 8×8 cell · `.art`

Standard C64 bitmap mode with 320×200 resolution. Each 8×8 character cell can use 2 colors from the 16-color palette.

BEST FOR High detail images, line art, text

- **Dithering:** Error diffusion recommended. Floyd-Steinberg or Atkinson work well.
- **Color Matching:** Perceptive gives natural results, Luma preserves brightness detail.

Multicolor (160×200)

160×200 · 4 per 4×8 cell · `.kla`

C64 multicolor bitmap with double-wide pixels. Each 4×8 character cell can use 4 colors: 3 from the palette + 1 shared background color.

BEST FOR Colorful images, photos, artwork with many color areas

- **Pixel Merge:** Average: Blends adjacent pixels for smoother results. Brightest: Picks the brighter pixel, preserves highlights.
- **Dithering:** Ordered dithering (Bayer) often works better than error diffusion in multicolor mode.

PETSCII (40×25)

40×25 characters · 16 (2 per cell) · `.prg`

Character-based graphics using the C64's built-in PETSCII character set. 40×25 character cells (320×200 pixels). Each 8×8 cell displays one of 256 PETSCII characters with 2 colors: global background + per-cell foreground.

BEST FOR ASCII/ANSI-style art, BBS graphics, text-based imagery, retro terminal aesthetics

- **Pattern Matching:** XOR-based algorithm finds the best matching PETSCII character for each 8×8 tile.
- **Color Selection:** Background: most common dark color. Foreground: most contrasting color per cell.
- **Color Matching:** Perceptive matching recommended for best character/color selection.

Chapter 5 shows Pixel Merge on a Multicolor picture (Figure 5-6).

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Graphics Mode	HiRes (320×200), Multicolor (160×200), PETSCII (40×25)	Multicolor (160×200)	The graphics mode, described above.
Dithering	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue 8x8, Blue 16x16	Atkinson	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Pixel Merge	Average, Brightest	Average	How the two halves of a double-wide pixel are combined: Average blends them, Brightest keeps the brighter one.
Color Match	Euclidean, Perceptive, Luma, Chroma, Mahalanobis	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2.5	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.10 MSX

TMS9918 · V9938

INTRODUCED
1983

VIDEO HARDWARE
TMS9918 / V9938

COLOURS
16; 16 of 512; 256

NATIVE FILES
.sc2 .sc5 .sc8

SHORTCUT
⇧⌘0

MSX computers come in two generations of video chip. Screen 2 is the MSX1 mode of the TMS9918: 256×192 pixels, two colours in every 8-pixel stretch of a line, from a fixed palette of 16. Screens 5 and 8 belong to the MSX2's V9938. Screen 5 shows 16 colours chosen from 512, and Screen 8 shows 256 fixed colours. Files are written in BSAVE format (`.sc2`, `.sc5`, `.sc8`).

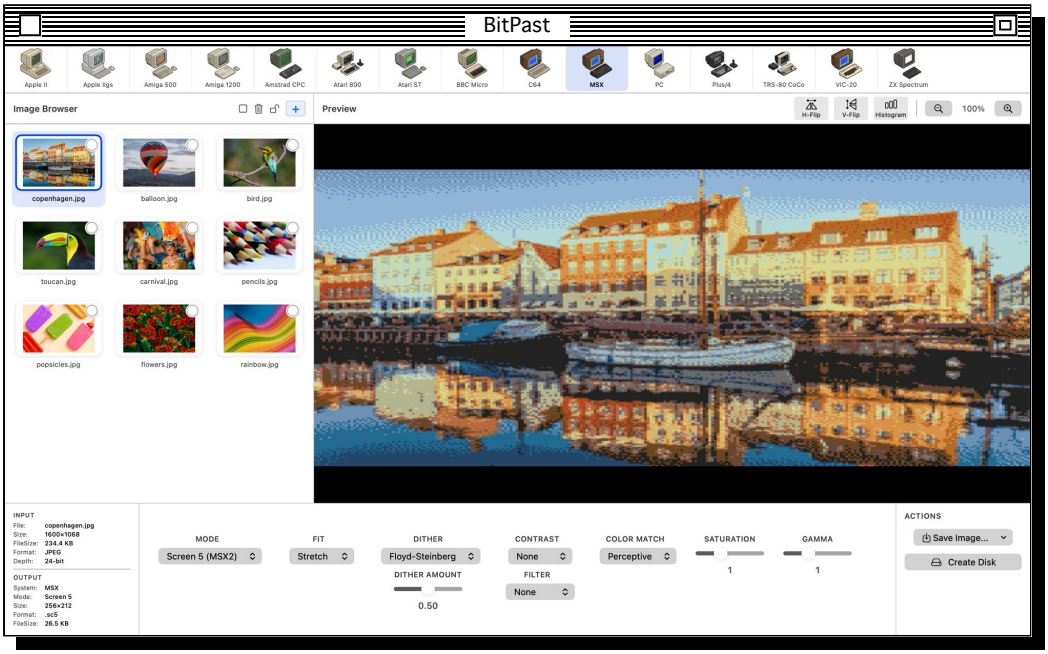
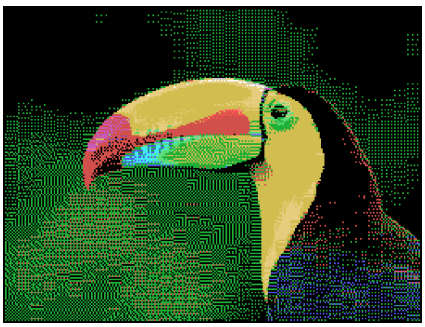


FIGURE 6-12
BitPast with the MSX selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
Screen 2 (MSX1)	256×192	2 per 8×1 stretch (16 fixed)	.sc2	13,063 bytes
Screen 5 (MSX2)	256×212	16 of 512	.sc5	27,175 bytes
Screen 8 (MSX2)	256×212	256 fixed (3-3-2 RGB)	.sc8	54,279 bytes



Screen 2



Screen 5



Screen 8

Screen 2 (MSX1)

256×192 · 2 per 8×1 stretch (16 fixed) · `.sc2`

MSX1 graphics mode using the TMS9918 video chip. 2 colors per 8×1 horizontal line from the fixed 16-color palette. Each 8-pixel horizontal segment has its own foreground and background color.

BEST FOR MSX1 game graphics, colorful images with horizontal color variation

- **Dithering:** Error diffusion works well within the 8×1 color constraints.
- **Color Matching:** Hue or Chroma recommended to preserve colors with the fixed TMS9918 palette.

Screen 5 (MSX2)

256×212 · 16 of 512 · `.sc5`

MSX2 graphics mode using the V9938 video chip. 16 colors selected from a 512-color palette (3 bits per channel). Similar to Atari ST graphics.

BEST FOR MSX2 game graphics, detailed images with custom palette

- **Dithering:** Floyd-Steinberg recommended for photos. Bayer for pixel art.
- **Color Matching:** Perceptive recommended. Optimal 16 colors are automatically selected from 512.

Screen 8 (MSX2)

256×212 · 256 fixed (3-3-2 RGB) · `.sc8`

MSX2 256-color mode with fixed 3-3-2 RGB palette (3 bits red, 3 bits green, 2 bits blue). No palette selection needed - colors are directly mapped.

BEST FOR Photographs, images with smooth gradients

- **Dithering:** Often not needed with 256 colors. Light dithering can help with banding.
- **Note:** Blue channel has only 4 levels (2 bits), so blue gradients may show more banding.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Mode	Screen 2 (MSX1), Screen 5 (MSX2), Screen 8 (MSX2)	Screen 5 (MSX2)	The graphics mode, described above.
Dither	None, Floyd-Steinberg, Atkinson, Bayer 2x2, Bayer 4x4, Bayer 8x8	Floyd-Steinberg	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma, Hue	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0.5 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.11 IBM PC

CGA · EGA · VGA · TEXT MODES

INTRODUCED
1981

VIDEO HARDWARE
CGA, EGA, VGA

COLOURS
4, 16 or 256

NATIVE FILES
.pcx .ans

SHORTCUT
⇧⌘P

The PC section covers the graphics standards that followed one another on DOS machines: CGA with four colours, EGA with 16, VGA with 256, and two text modes for ANSI art. Graphics modes are saved as PC Paintbrush (`.pcx`) pictures and text modes as ANSI files (`.ans`).

With the CGA mode, **CGA Palette** picks one of four fixed sets of three colours (plus black). With the EGA modes you can choose between the fixed 16 colours of the standard palette and 16 chosen from the full set of 64.

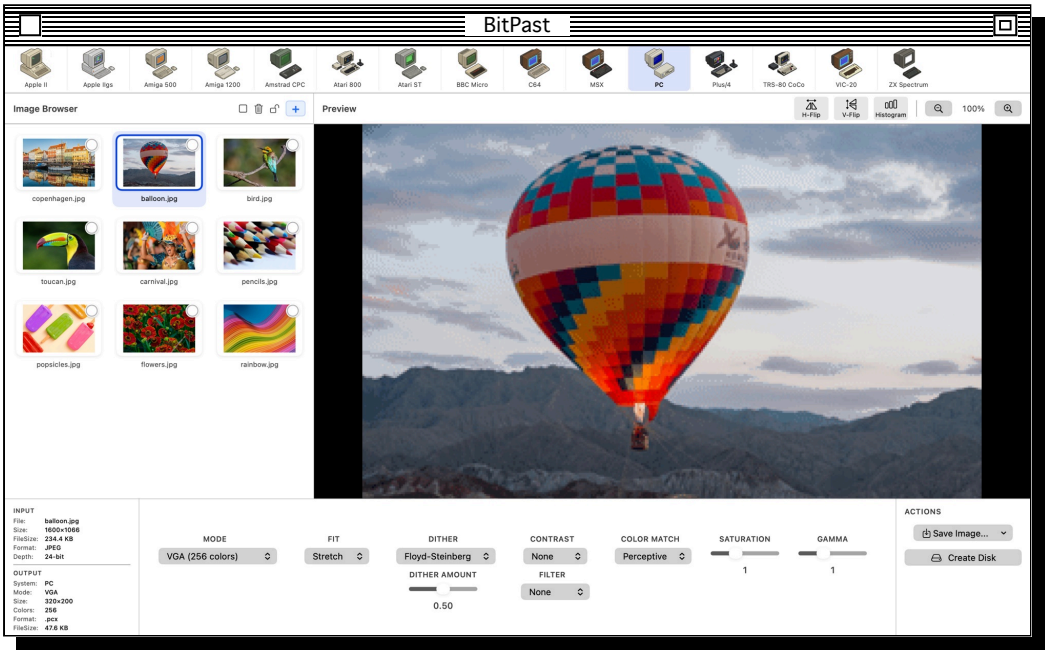
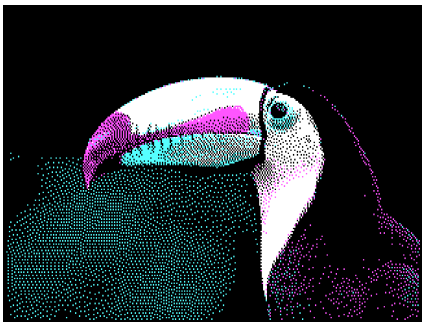


FIGURE 6-13
BitPast with the IBM PC selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
CGA (4 colors)	320×200	4 (fixed palettes)	.pcx	varies
EGA (16 colors)	320×200	16 (fixed)	.pcx	varies
EGA 64 (16 from 64)	320×200	16 of 64	.pcx	varies
VGA (256 colors)	320×200	256 of 262,144	.pcx	varies
CGA 80×25 Text	80×25 characters (640×200)	16 + 8 per cell	.ans	varies
VESA 132×50 Text	132×50 characters (1056×400)	16 + 8 per cell	.ans	varies



CGA



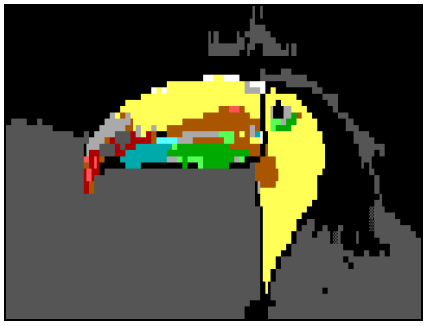
EGA



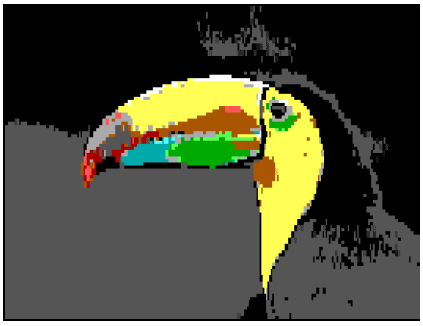
EGA 64



VGA



CGA 80×25 Text



VESA 132×50 Text

CGA (4 colors)

320×200 · 4 (fixed palettes) · .pcx

Color Graphics Adapter mode with 4 colors from fixed palettes. The iconic PC graphics standard from 1981.

BEST FOR Classic PC game graphics, retro DOS aesthetics

- **CGA Palette:** Cyan/Magenta/White (high intensity), Cyan/Magenta/Gray (low), Green/Red/Yellow (high), Green/Red/Brown (low)
- **Dithering:** Floyd-Steinberg strongly recommended due to limited 4-color palette.

EGA (16 colors)

320×200 · 16 (fixed) · .pcx

Enhanced Graphics Adapter mode using the standard fixed 16-color EGA palette. The same colors as CGA's 16-color text mode palette.

BEST FOR Authentic EGA look, classic DOS game graphics

- **Dithering:** Ordered dithering (Bayer) often produces good results. 16 colors allow reasonable color reproduction.
- **Color Matching:** Hue or Chroma recommended to preserve colors with this fixed palette. Perceptive may map saturated colors to grays.

EGA 64 (16 from 64)

320×200 · 16 of 64 · .pcx

Enhanced Graphics Adapter mode with 16 colors selected from the full 64-color EGA palette. Allows optimal color selection per image.

BEST FOR Images that benefit from custom color selection, Sierra adventure games aesthetic

- **Dithering:** Ordered dithering (Bayer) often produces good results.
- **Color Matching:** Perceptive recommended. Optimal 16 colors are automatically selected from the 64-color EGA palette.

VGA (256 colors)

320×200 · 256 of 262,144 · .pcx

The legendary VGA Mode 13h with 256 colors from the full 262,144-color VGA palette. The standard for DOS games from 1987-1997.

BEST FOR DOS game graphics, pixel art, photographs

- **Dithering:** Often not needed with 256 colors. Use sparingly for photographs.
- **Color Matching:** Uses adaptive palette generation (median cut) for optimal 256-color palette.

CGA 80×25 Text

80×25 characters (640×200) · 16 + 8 per cell · `.ans`

Classic DOS text mode using 8×8 character cells. 80 columns × 25 rows with 16 foreground and 8 background colors. Resolution: 640×200 pixels.

BEST FOR ASCII art, BBS graphics, ANSI art aesthetics

- **Characters:** Uses block characters and shading patterns to approximate image content
- **Colors:** 16 foreground colors, 8 background colors per character cell

VESA 132×50 Text

132×50 characters (1056×400) · 16 + 8 per cell · `.ans`

Extended VESA text mode with 132 columns × 50 rows. Higher resolution text display (1056×400 pixels) for more detailed ASCII art.

BEST FOR High-detail ASCII art, detailed text mode graphics

- **Characters:** Same block characters as CGA text mode but with more cells for finer detail
- **Resolution:** 132×50 characters = 1056×400 pixels

CGA palettes

CGA PALETTE	COLOURS (PLUS BLACK)
Cyan/Magenta/White	Cyan, magenta and white, in high intensity.
Cyan/Magenta/Gray	Cyan, magenta and light grey, in low intensity.
Green/Red/Yellow	Light green, light red and yellow, in high intensity.
Green/Red/Brown	Green, red and brown, in low intensity.

Chapter 5 shows how Color Match affects the fixed EGA palette (Figure 5-5).

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Mode	CGA (4 colors), EGA (16 colors), EGA 64 (16 from 64), VGA (256 colors), CGA 80×25 Text, VESA 132×50 Text	VGA (256 colors)	The graphics mode, described above.
CGA Palette	Cyan/Magenta/White, Cyan/Magenta/Gray, Green/Red/Yellow, Green/Red/Brown	Cyan/Magenta/White	The fixed set of three colours (plus black) used in CGA mode.
Dither	None, Floyd-Steinberg, Atkinson, Bayer 2x2, Bayer 4x4, Bayer 8x8	Floyd-Steinberg	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma, Hue	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0.5 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.12 Commodore Plus/4

TED · 128-COLOUR PALETTE

INTRODUCED
1984

VIDEO HARDWARE
TED

COLOURS
128 (16 hues × 8 levels)

NATIVE FILES
.prg

SHORTCUT
⇧⌘L

The Plus/4's TED chip has a palette of 128 colours: 16 hues, each in 8 levels of brightness. Like the Commodore 64 it works in cells: 8×8 cells with two colours in HiRes mode, and 4×8 cells with four colours in Multicolor mode — but with far more colours to choose from. Pictures are saved as `.prg` programs.

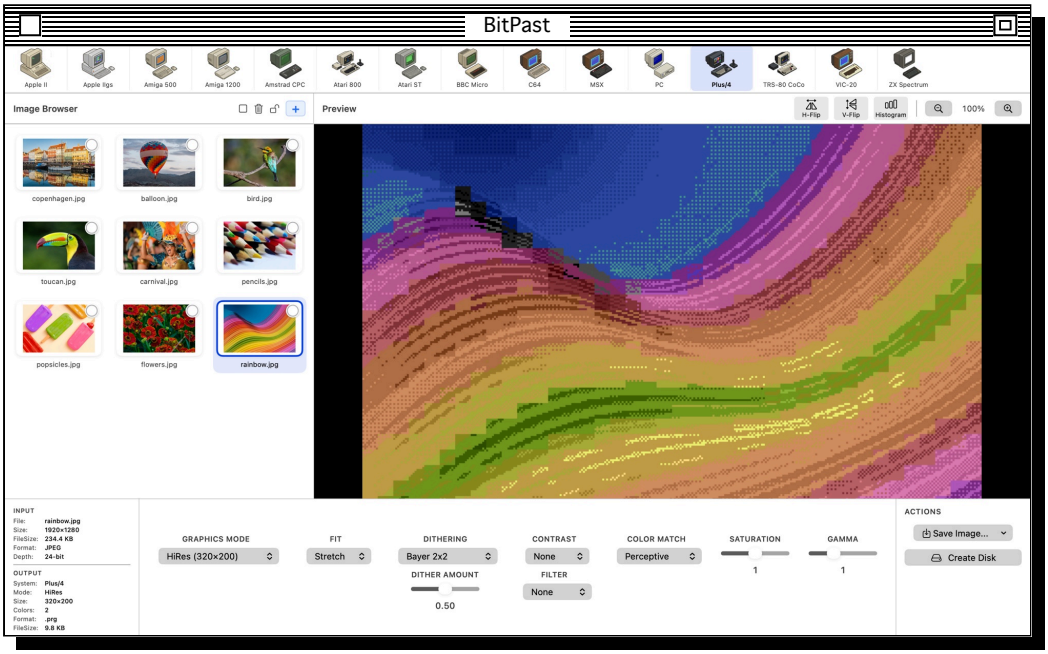
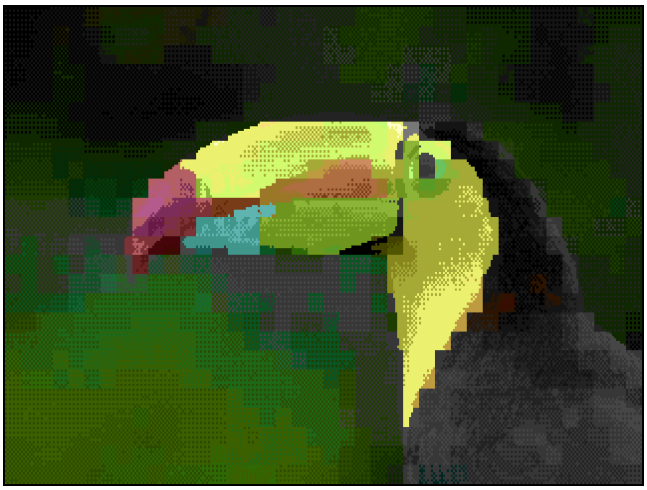


FIGURE 6-14
BitPast with the Commodore Plus/4 selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
HiRes (320×200)	320×200	2 per 8×8 cell (of 128)	`.prg`	10,000 bytes
Multicolor (160×200)	160×200	4 per 4×8 cell (of 128)	`.prg`	10,002 bytes



HiRes



Multicolor

HiRes (320×200)

320×200 · 2 per 8×8 cell (of 128) · `.prg`

Plus/4 high-resolution bitmap mode with 320×200 pixels. Each 8×8 character cell can use 2 colors from the 128-color TED palette.

BEST FOR Detailed images, line art, high resolution needs

- **Dithering:** Error diffusion works well. Floyd-Steinberg recommended.
- **Color Matching:** The large 128-color palette provides excellent color matching. Perceptive recommended.

Multicolor (160×200)

160×200 · 4 per 4×8 cell (of 128) · `.prg`

Plus/4 multicolor bitmap with double-wide pixels. Each 4×8 character cell can use 4 colors: 2 global colors + 2 per cell from the 128-color palette.

BEST FOR Colorful photos, artwork with smooth gradients

- **Pixel Merge:** Average: Blends adjacent pixels smoothly. Brightest: Preserves highlight detail.
- **Dithering:** Both error diffusion and ordered dithering work well with the large palette.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Graphics Mode	HiRes (320×200), Multicolor (160×200)	HiRes (320×200)	The graphics mode, described above.
Dithering	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue 8x8, Blue 16x16	Bayer 2x2	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Pixel Merge	Average, Brightest	Average	How the two halves of a double-wide pixel are combined: Average blends them, Brightest keeps the brighter one.
Color Match	Euclidean, Perceptive, Luma, Chroma, Mahalanobis	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.13 TRS-80 Color Computer

MC6847 VDG · GIME

INTRODUCED
1980

VIDEO HARDWARE
MC6847 and GIME

COLOURS
2, 4 or 16

NATIVE FILES
.bin .cm3

SHORTCUT
⇧⌘T

The Color Computer's original video chip, the MC6847, works with a handful of fixed colours in modes the BASIC programmer knows as PMODE 1 to 4. The CoCo 3 added the GIME chip, with 320×200 pictures in 16 colours and 640×200 pictures in 4, chosen from a palette of 64.

For the four-colour PMODEs, **Color Set** picks one of two fixed sets of colours, or the NTSC artifact colours that appear on a composite monitor with the high-resolution two-colour mode.

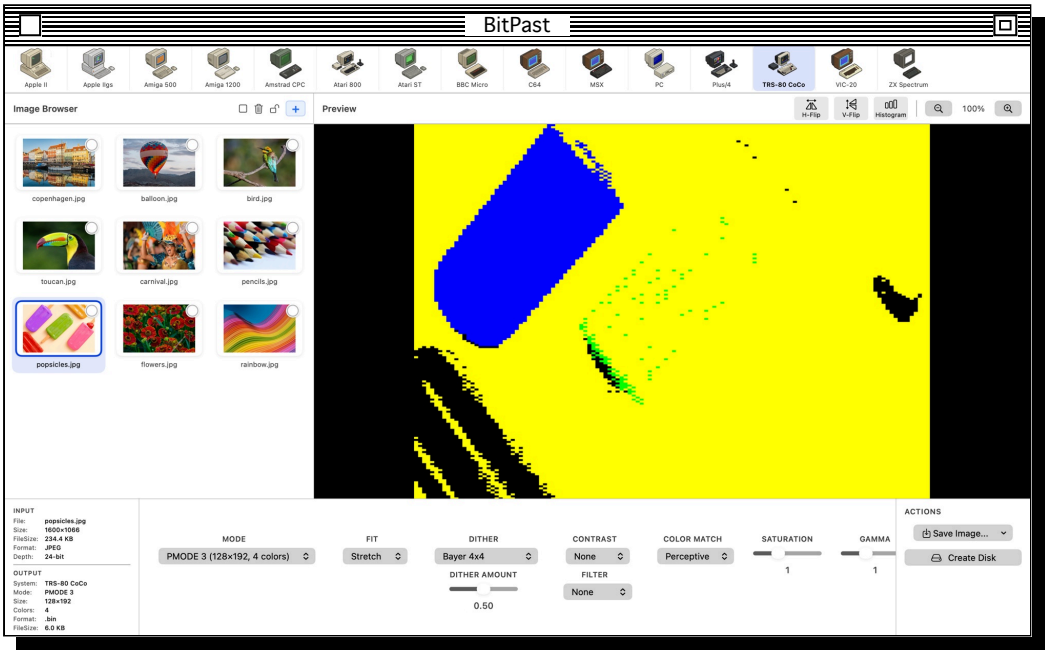
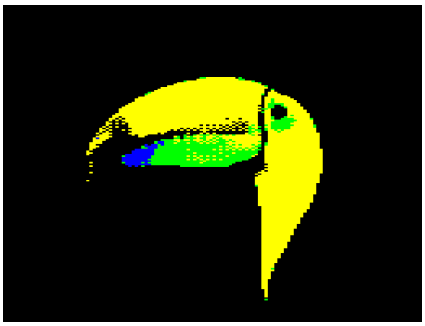


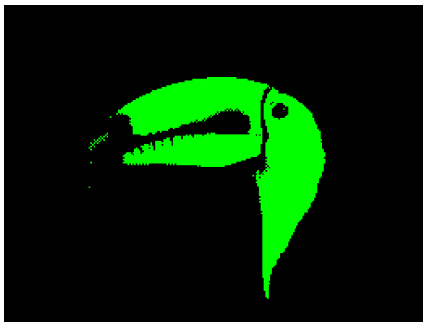
FIGURE 6-15
BitPast with the TRS-80 Color Computer selected.

Modes

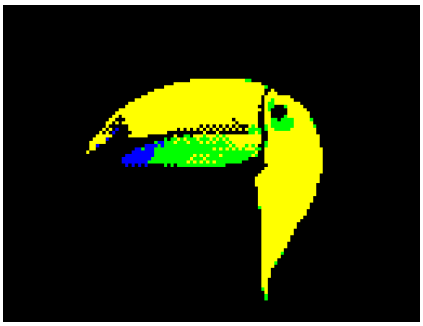
MODE	SCREEN	COLOURS	FILE	SIZE
PMODE 3 (128×192, 4 colors)	128×192	4 (colour set)	.bin	6,144 bytes
PMODE 4 (256×192, 2 colors)	256×192	2 (+ artifact)	.bin	6,144 bytes
PMODE 1 (128×96, 4 colors)	128×96	4 (colour set)	.bin	3,072 bytes
PMODE 2 (128×192, 2 colors)	128×192	2	.bin	3,072 bytes
CoCo 3 (320×200, 16 colors)	320×200	16 of 64	.cm3	32,000 bytes
CoCo 3 (640×200, 4 colors)	640×200	4 of 64	.cm3	32,000 bytes



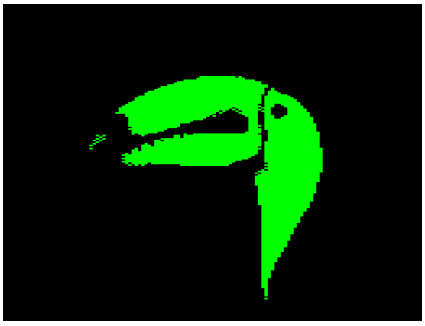
PMODE 3



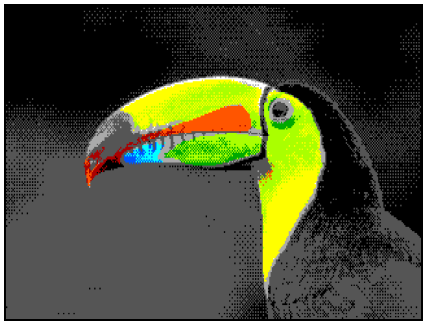
PMODE 4



PMODE 1



PMODE 2



CoCo 3 320×200



CoCo 3 640×200

PMODE 3 (128×192, 4 colors)

128×192 · 4 (colour set) · `.bin`

Standard CoCo graphics mode using the MC6847 VDG chip. 128×192 resolution with 4 colors from either Color Set 0 or Set 1. Most common mode for CoCo 1/2 graphics.

BEST FOR General CoCo graphics, games, colorful images

- **Color Set 0:** Green, Yellow, Blue, Red on black background
- **Color Set 1:** Buff, Cyan, Magenta, Orange on black background
- **Dithering:** Bayer 4×4 recommended. Floyd-Steinberg for photos.

PMODE 4 (256×192, 2 colors)

256×192 · 2 (+ artifact) · `.bin`

High-resolution CoCo mode with 256×192 pixels but only 2 colors. Can use NTSC artifact coloring for additional colors on real hardware.

BEST FOR Line art, text, high contrast images, artifact color graphics

- **Artifact (NTSC):** Enables NTSC artifact colors (Blue/Orange) for additional color variety on composite displays
- **Dithering:** Bayer dithering creates good patterns in 2-color mode

PMODE 1 (128×96, 4 colors)

128×96 · 4 (colour set) · `.bin`

Low-resolution CoCo mode with 128×96 pixels and 4 colors. Uses less memory than other modes. Good for simple graphics.

BEST FOR Simple graphics, memory-constrained applications

- **Color Set:** Same color sets as PMODE 3
- **Dithering:** Light dithering recommended due to low resolution

PMODE 2 (128×192, 2 colors)

128×192 · 2 · `.bin`

Medium-resolution mode with 128×192 pixels and 2 colors. Uses half the memory of PMODE 3.

BEST FOR Monochrome graphics, line art when memory is limited

- **Dithering:** Bayer dithering recommended for photos

CoCo 3 (320×200, 16 colors)

320×200 · 16 of 64 · `.cm3`

CoCo 3 GIME mode with 320×200 resolution and 16 colors from the 64-color RGB palette (2 bits per channel). Requires CoCo 3 hardware.

BEST FOR Detailed colorful graphics, photos, CoCo 3 games

- **Fixed Palette:** Uses a standard 16-color palette for emulator compatibility
- **Dithering:** Floyd-Steinberg recommended for photos, Bayer for pixel art

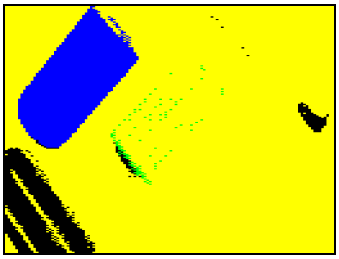
CoCo 3 (640×200, 4 colors)

640×200 · 4 of 64 · `.cm3`

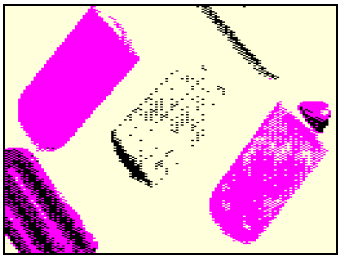
CoCo 3 high-resolution mode with 640×200 pixels and 4 colors. Good balance of resolution and color for text and detailed graphics.

BEST FOR Text, CAD, detailed line art, desktop applications

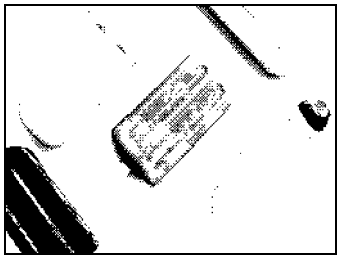
- **Dithering:** Ordered dithering (Bayer) works well with limited 4-color palette



Set 0
Green, yellow, blue, red.



Set 1
Buff, cyan, magenta, orange.



Artifact (NTSC)
PMODE 4 with artifact colours.

FIGURE 6-16
The colour sets of the original CoCo.

Color Set applies to PMODE 3 and PMODE 1. **Artifact (NTSC)** is meant for PMODE 4: on a composite monitor the two colours of that mode produce blue and orange fringes, which BitPast uses to give the picture additional colour.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Mode	PMODE 3 (128×192, 4 colors), PMODE 4 (256×192, 2 colors), PMODE 1 (128×96, 4 colors), PMODE 2 (128×192, 2 colors), CoCo 3 (320×200, 16 colors), CoCo 3 (640×200, 4 colors)	PMODE 3 (128×192, 4 colors)	The graphics mode, described above.
Color Set	Set 0 (Green/Yellow/Blue/Red), Set 1 (Buff/Cyan/Magenta/Orange), Artifact (NTSC)	Set 0 (Green/Yellow/Blue/Red)	The set of colours used by the CoCo's four-colour modes.
Dither	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue Noise 8x8, Blue Noise 16x16	Bayer 4x4	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma, Hue	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0.5 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.14 VIC-20

VIC · 16 COLOURS

INTRODUCED
1980

VIDEO HARDWARE
VIC

COLOURS
16

NATIVE FILES
.prg

SHORTCUT
⇧⌘V

The VIC-20 builds its pictures from a custom character set: the screen is 22 by 23 characters, each 8 pixels high. HiRes mode gives two colours per 8×8 cell at 176×184 pixels. LowRes is the VIC-20's multicolor mode, with double-wide pixels (88×184) and four colours per cell: background, border, auxiliary and one chosen for the cell. Files are `.prg` programs that show the picture when you run them.

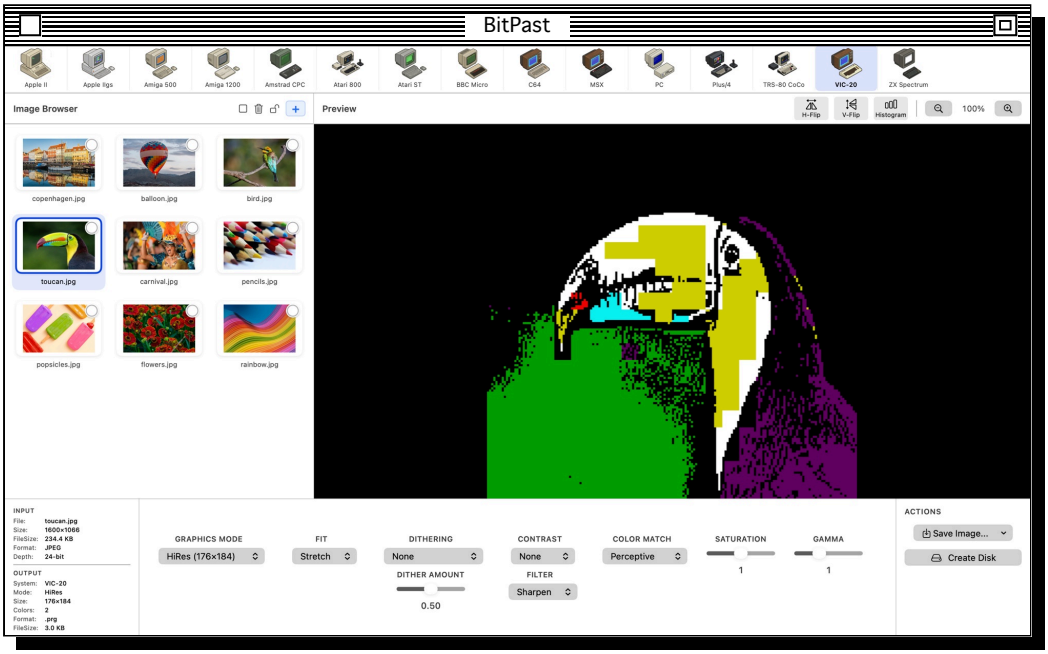


FIGURE 6-17
BitPast with the VIC-20 selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
HiRes (176×184)	176×184	2 per 8×8 cell	`.prg`	varies
LowRes (88×184)	88×184	4 per cell	`.prg`	varies



HiRes



LowRes

HiRes (176×184)

176×184 · 2 per 8×8 cell · `.prg`

VIC-20 high-resolution mode using custom character set. 22×23 character cells, each 8×8 pixels with 2 colors per cell.

BEST FOR Detailed images, line art, higher resolution needs

- **Dithering:** Error diffusion works well with the limited 2 colors per cell.
- **Color Matching:** Perceptive recommended for natural color selection.

LowRes (88×184)

88×184 · 4 per cell · `.prg`

VIC-20 multicolor mode with double-wide pixels. 22×23 cells, each 4×8 pixels with 4 colors: background, border, auxiliary + 1 per cell.

BEST FOR Colorful images, photos, artwork needing more colors per area

- **Pixel Merge:** Average: Blends pixels for smoother look. Brightest: Preserves highlight detail.
- **Dithering:** Ordered dithering often works better than error diffusion.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Graphics Mode	HiRes (176×184), LowRes (88×184)	HiRes (176×184)	The graphics mode, described above.
Dithering	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue 8x8, Blue 16x16	None	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	Sharpen	A quick filter applied before converting.
Pixel Merge	Average, Brightest	Average	How the two halves of a double-wide pixel are combined: Average blends them, Brightest keeps the brighter one.
Color Match	Euclidean, Perceptive, Luma, Chroma, Mahalanobis	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

6.15 ZX Spectrum

ULA · 15 COLOURS

INTRODUCED
1982

VIDEO HARDWARE
ULA

COLOURS
15

NATIVE FILES
.scr

SHORTCUT
⇧⌘Z

The ZX Spectrum has one graphics mode: 256×192 pixels in which every 8×8 cell may use only two colours, called *ink* and *paper*, taken from 8 colours in two levels of brightness (15 colours in all, because bright black is the same as black). Where a picture needs more colours in a cell than the hardware allows, the colours of neighbouring cells "clash", and BitPast's output shows that clash exactly as the real machine would. Files are `.scr` screen dumps of 6,912 bytes: 6,144 bytes of bitmap and 768 of colour attributes.

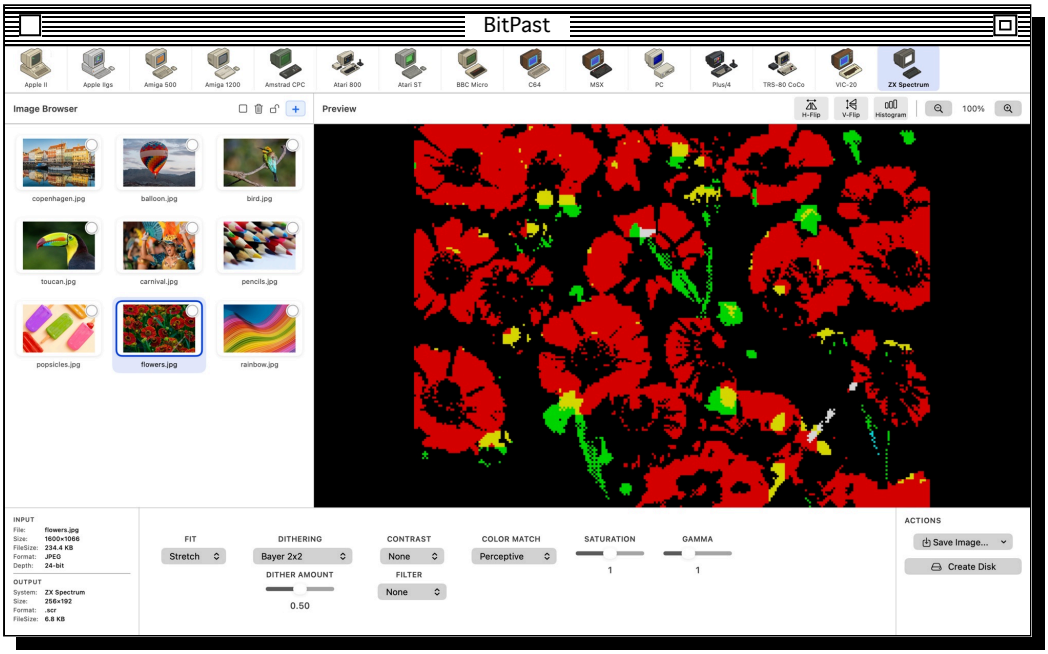


FIGURE 6-18
BitPast with the ZX Spectrum selected.

Modes

MODE	SCREEN	COLOURS	FILE	SIZE
Standard (256×192)	256×192	2 per 8×8 cell (15 colours)	`.scr`	6,912 bytes



Standard (256×192)

Standard (256×192)

256×192 · 2 per 8×8 cell (15 colours) · `.scr`

ZX Spectrum screen with 256×192 pixels. Each 8×8 attribute cell has 2 colors (ink + paper) from the 15-color palette. Famous for 'attribute clash' when colors cross cell boundaries.

BEST FOR Authentic Spectrum look, pixel art, images with clearly defined color regions

- **Dithering:** Works within attribute constraints. Floyd-Steinberg or Atkinson recommended.
- **Contrast:** CLAHE or SWAHE can help improve detail in attribute-constrained images.
- **Color Matching:** Perceptive gives natural results within the limited palette.

Settings (every system also offers Fit and Aspect, Chapter 4)

SETTING	CHOICES	DEFAULT	WHAT IT DOES
Dithering	None, Floyd-Steinberg, Atkinson, Noise, Bayer 2x2, Bayer 4x4, Bayer 8x8, Bayer 16x16, Blue 8x8, Blue 16x16	Bayer 2x2	How colours that the machine can't show are approximated (Chapter 5).
Dither Amount	slider, 0 to 1	0.50	How strongly the dither is applied; 0 is none, 1 is full strength.
Contrast	None, HE, CLAHE, SWAHE	None	Contrast enhancement applied before the colours are chosen.
Filter	None, Lowpass, Sharpen, Emboss, Edge	None	A quick filter applied before converting.
Color Match	Euclidean, Perceptive, Luma, Chroma, Mahalanobis	Perceptive	How BitPast judges which colour is closest.
Saturation	slider, 0 to 2	1	Colour intensity; 1 leaves it unchanged.
Gamma	slider, 0.5 to 2	1	Brightness of the mid-tones; below 1 brightens, above 1 darkens.

CHAPTER 7

7 The Palette Editor

Changing the colours of an Apple IIgs picture

An Apple IIgs picture is stored together with its palettes, and the palette editor lets you change them: make a sky bluer, a skin tone warmer, or fix one stubborn line. It is available for the Apple IIgs only.

7.1 Opening the editor

Select the Apple IIgs, choose a picture in a mode that has palettes, and click Palette above the preview. The editor opens in its own window. The button is dimmed until a picture has been converted.

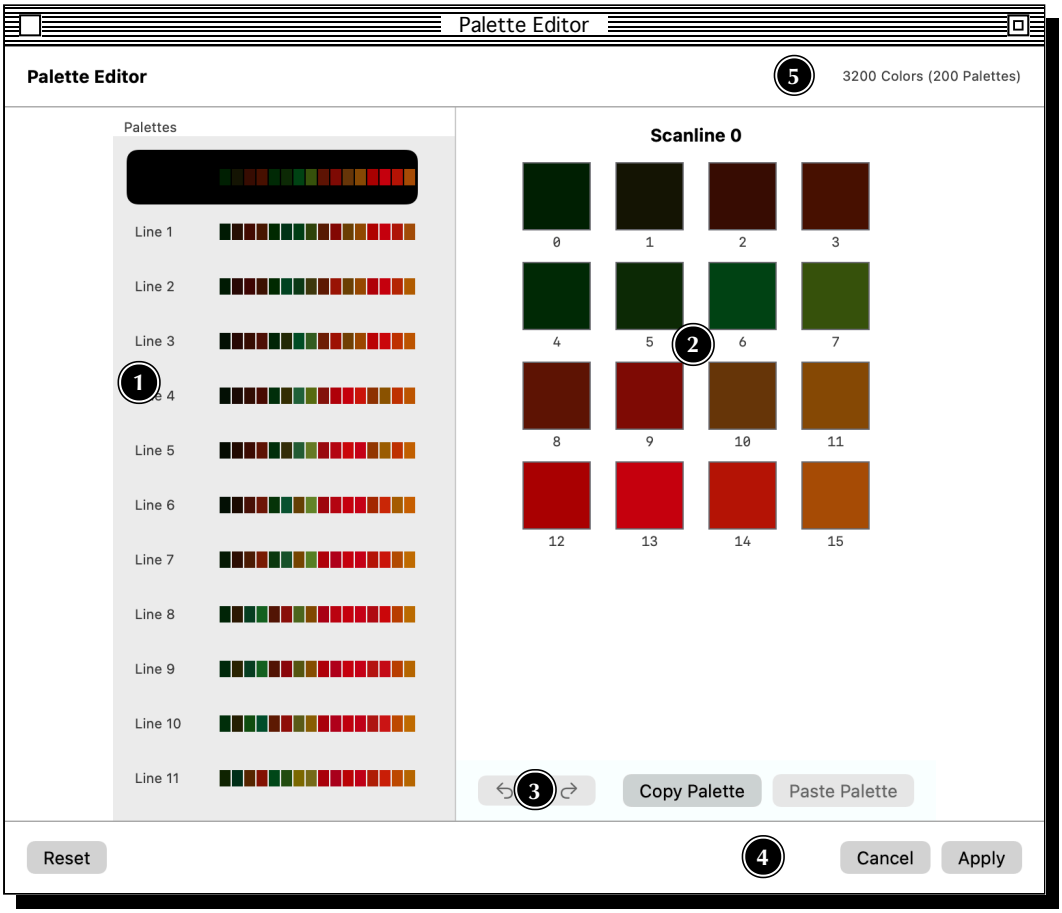


FIGURE 7-1
The palette editor with a 3,200-colour picture.

1. **Palette list.** One entry for each palette, with a strip showing its sixteen colours. In 3,200-colour mode the entries are lines (Line 0 to Line 199); in 256-colour mode they are palettes (Palette 0 to Palette 15).

2. **Colour grid.** The sixteen colours of the selected palette.
3. **Undo, redo, Copy Palette and Paste Palette.**
4. **Reset, Cancel and Apply.**
5. **Title.** The editor tells you the mode: for example, 3200 Colors (200 Palettes).

7.2 Changing a colour

1. Click an entry in the palette list. The grid shows its sixteen colours.
2. Click a colour in the grid. The standard macOS colour panel opens; choose the colour you want. The colour's number, its hexadecimal value and its IIgs value ($RGB) are shown below the grid.
3. Repeat for any other colours. Undo and Redo (⌘Z and ⇧⌘Z) step through your changes.
4. Click Apply to keep the changes, Cancel to leave without them, or Reset to put the palettes back as BitPast made them.

The IIgs can show only 4,096 colours: sixteen levels each of red, green and blue. A colour you choose is rounded to the nearest of these when it is written to the file, which is why the editor also shows the IIgs value.

7.3 Copying palettes

Copy Palette copies the selected palette; Paste Palette puts it into another entry of the list. This is the quickest way to give a run of lines the same colours, or to move a good palette to a place that needs it.

7.4 What Apply does

When you click Apply, the preview is redrawn with your colours at once. The edits are also written into the picture's file — the `.shr` or `.3200` — so Save Image…, batch export and Create Disk all use the edited palettes for that picture.

IMPORTANT Changing any setting, or moving or flipping the picture, converts it again from the original and discards palette edits. Make palette edits last, once everything else is as you want it.

Edits belong to the picture and the system they were made for. If you switch to another system and back, convert again.

CHAPTER 8

8 Saving and Exporting

Pictures and the machine's files

BitPast can save a picture in two ways: as an ordinary picture file for viewing and sharing, and as the machine's own file for loading on the machine. It can also do either for many pictures at once.

8.1 Saving one picture

Click Save Image… in the Actions column and choose a format:

CHOICE	WHAT YOU GET
PNG	The preview as an ordinary picture, without loss. The best choice for sharing the result.
JPG	The preview as a JPEG. Smaller, but its compression softens the sharp pixels.
GIF	The preview as a GIF.
TIFF	The preview as a TIFF.
Native Format (.ext)	The machine's own file. The menu shows the extension for the selected mode, for example `.kla`.

TIP The preview is small, because it is the machine's own screen. To enlarge a saved PNG without blurring it, open it in an app that can scale with nearest-neighbour interpolation, or scale it by a whole number such as 3 or 4.

8.2 Native files

A native file contains the screen exactly as the machine's hardware stores it. What is in it depends on the machine. A Spectrum `.scr` holds the bitmap and the colour attributes. An Apple IIgs `.shr` holds the 200 lines of bitmap, a scan control byte for each line, and the palettes. A Commodore 64 `.kla` holds the bitmap, the screen and colour memory and the background colour.

Appendix A lists the file for every mode, with its size, and describes the layout of several of them. Chapter 9 explains how to put native files onto a disk image.

8.3 Saving several pictures at once

1. Select the pictures with the small circles in the Image Browser (Chapter 4), or use the checkbox above the browser to select all of them.

2 Click the Save Image... menu. It now reads Save 3 Images..., or however many you selected.

3 Choose a format. BitPast asks you to choose a folder, and converts and saves every selected picture in it, showing its progress as Exporting 2/3....

Each picture is converted with the current settings, or with its own settings if you locked them (Chapter 4). Each file is named after its picture, with the extension of the format. The position of each picture, and any flip, is used too.

CHAPTER 9

9 Disk Images

Putting your pictures on a disk

A disk image is a single file that holds the contents of a whole floppy disk. Emulators load disk images directly, and floppy emulators and disk-copying tools can put them on real media. BitPast can create disk images for all fifteen systems and fill them with your converted pictures.

9.1 Creating a disk image

1. Add your pictures and choose the system. Adjust the settings until the pictures look right.
2. Select the pictures that should go on the disk. If you select none, every picture in the browser is used.
3. Click Create Disk in the Actions column. A sheet opens with the system already chosen to match the main window.
4. Type a volume name, and choose the disk format and size. The sheet tells you how many pictures will be converted and added.
5. Click Create Disk Image and choose where to save it. BitPast converts every picture, writes the files onto the disk and shows its progress.

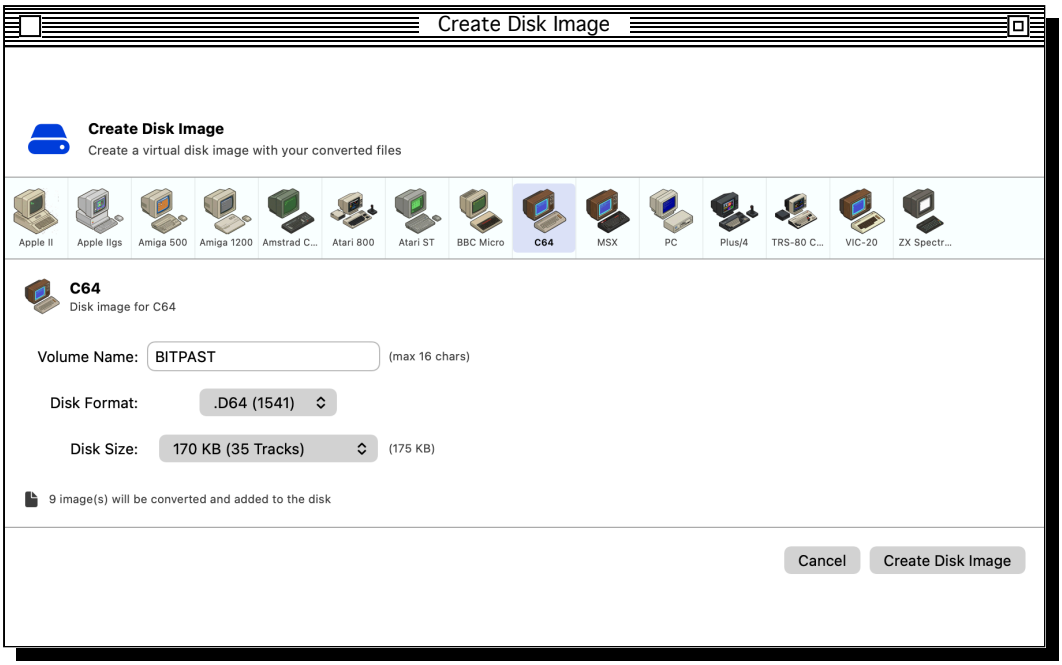


FIGURE 9-1
The Create Disk Image sheet, with the Commodore 64 chosen.

TIP Leave the sheet on the system that is selected in the main window. The pictures are converted for that system, and the disk should be made for the same one.

9.2 Disk formats

SYSTEM	IMAGE TYPE	SIZES	FILE SYSTEM	EMULATORS
Apple II, Apple IIgs	`.PO`, `.2MG`, `.HDV`	140 KB, 800 KB, 32 MB	ProDOS	AppleWin, KEGS, GSport, Virtual II
Commodore 64, VIC-20, Plus/4	`.D64`, `.D71`, `.D81`	170 KB, 340 KB, 800 KB	CBM DOS (files are PRG type)	VICE (x64, x128, xvic, xplus4)
Amiga 500, Amiga 1200	`.ADF`	880 KB, 1.76 MB	OFS (Original File System)	FS-UAE, WinUAE, Amiberry
Atari 800	`.ATR`	90 KB, 130 KB, 180 KB, 360 KB	Atari DOS 2.0S; DOS 2.5 for 130 KB	Altirra, Atari800
Atari ST	`.ST`	360 KB, 720 KB, 1.44 MB	FAT12	Hatari, Steem
BBC Micro	`.SSD`, `.DSD`	100 KB, 200 KB, 400 KB	Acorn DFS (up to 31 files)	BeebEm, b-em, JSBeeb
MSX	`.DSK`	360 KB, 720 KB	MSX-DOS (FAT12)	openMSX, blueMSX, fMSX
Amstrad CPC	`.DSK`	180 KB, 360 KB	AMSDOS (extended DSK)	WinAPE, Arnold, Caprice32
ZX Spectrum	`.TRD`, `.DSK`	640 KB, 180 KB	TR-DOS (80 tracks, double-sided); +3DOS	Fuse, ZX Spin, SpecEmu
IBM PC	`.IMG`	360 KB, 720 KB, 1.2 MB, 1.44 MB	FAT12 / FAT16	DOSBox, PCem, 86Box
TRS-80 CoCo	`.DSK`	180 KB, 360 KB	RS-DOS (360 KB as double-sided JVC)	CoCo emulators that read JVC disks

The Commodore disks: D64 is a 1541 disk (35 tracks), D71 a 1571 disk (70 tracks) and D81 a 1581 disk (80 tracks). The Amiga's 880 KB image is a double-density disk; 1.76 MB is high density. The Apple's 140 KB image is a 5.25-inch disk, 800 KB a 3.5-inch disk, and 32 MB a hard-disk image. The BBC's SSD is single-sided and DSD double-sided.

9.3 Volume names

Every system has its own rules for the name of a disk, and the sheet tells you the longest name allowed. BitPast corrects names that break the rules.

SYSTEM	RULE
Apple II, Apple IIgs	Up to 15 characters: letters, digits and periods, starting with a letter.
Commodore	Up to 16 characters; most printable PETSCII characters.
Amiga	Up to 30 characters; most ASCII characters except : and /.
Atari, PC, MSX	8 to 11 characters, depending on the system: letters and digits only.
BBC Micro	Up to 12 characters, letters and digits.

9.4 Choosing a size

Choose a disk large enough for all your pictures. A picture that does not fit is left off the disk rather than being written as a damaged file, so if a picture is missing, make the disk larger.

A native file is often much smaller than a disk — a Spectrum picture is under 7 KB — so a single disk usually holds many pictures. Appendix A gives the size of every file.

CHAPTER 10

10 Appearance and Help

Looks, shortcuts and the built-in guide

BitPast can look like a modern Mac app, or like one of three classic computers. This chapter also lists the keyboard shortcuts and describes the guide built into the app.

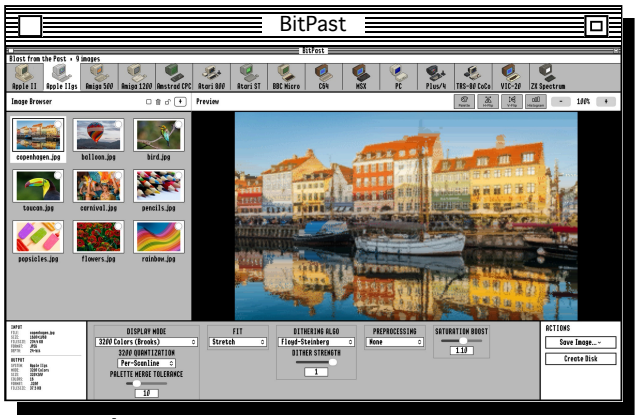
10.1 Appearance

Choose BitPast ▸ Settings… (⌘,) to open the Settings window. It has one control, Appearance, with six choices:

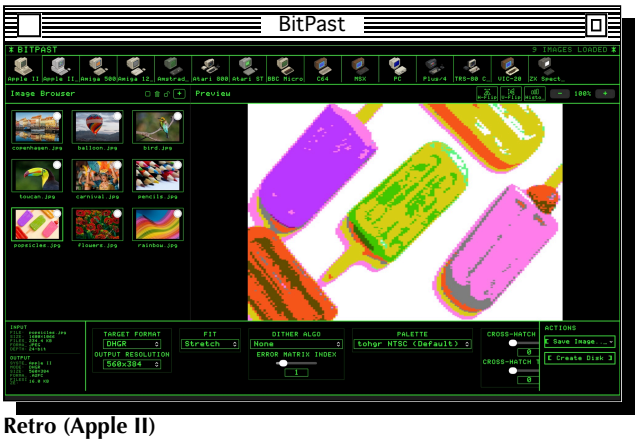
CHOICE	WHAT YOU GET
System	Follows the light or dark setting of your Mac.
Light	Always light.
Dark	Always dark.
Retro (Apple IIgs)	The interface drawn in the style of the IIgs's GS/OS desktop: platinum grey, with the system's bitmap type.
Retro (Apple II)	A green-phosphor monochrome screen, with the Apple II's character set.
Retro (Commodore 64)	The C64's blue screen, with its character set.



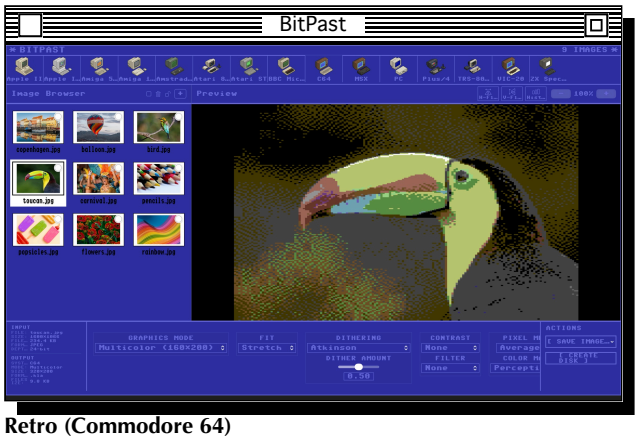
Dark



Retro (Apple IIgs)



Retro (Apple II)



Retro (Commodore 64)

FIGURE 10-1

BitPast in four of its six appearances. (Light is the look shown in the rest of this guide.)

The retro appearances change only how BitPast looks. They do not change what it does.

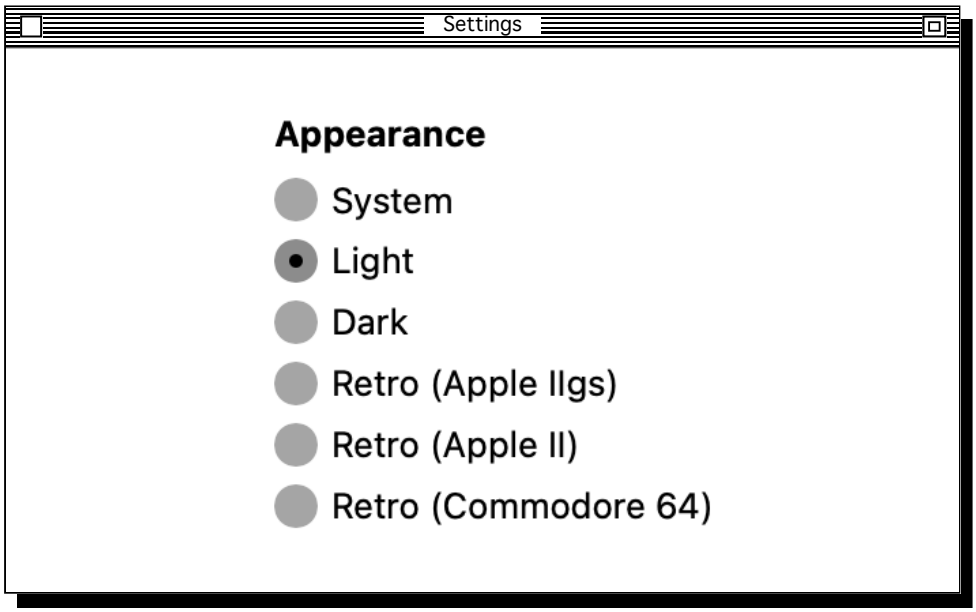


FIGURE 10-2

The Settings window.

10.2 Keyboard shortcuts

SYSTEM	SHORTCUT	SYSTEM	SHORTCUT
Apple II	⇧⌘1	**Commodore 64**	⇧⌘9
Apple IIgs	⇧⌘2	**MSX**	⇧⌘0
Amiga 500	⇧⌘3	**IBM PC**	⇧⌘P
Amiga 1200	⇧⌘4	**Commodore Plus/4**	⇧⌘L
Amstrad CPC	⇧⌘5	**TRS-80 Color Computer**	⇧⌘T
Atari 800	⇧⌘6	**VIC-20**	⇧⌘V
Atari ST	⇧⌘7	**ZX Spectrum**	⇧⌘Z
BBC Micro	⇧⌘8		

COMMAND	SHORTCUT
Graphics Mode Guide	⌘?
Settings	⌘,
Undo (palette editor)	⌘Z
Redo (palette editor)	⇧⌘Z
Select a range of pictures	Shift-click the second picture
Centre the picture in the preview	Double-click the preview

10.3 The Graphics Mode Guide

Choose Help ▸ Graphics Mode Guide (⌘?) to open a window that describes every system, mode and setting. Systems are listed on the left, and General Options, Preprocessing Filters and Disk Images under Reference. Click one to read about it.

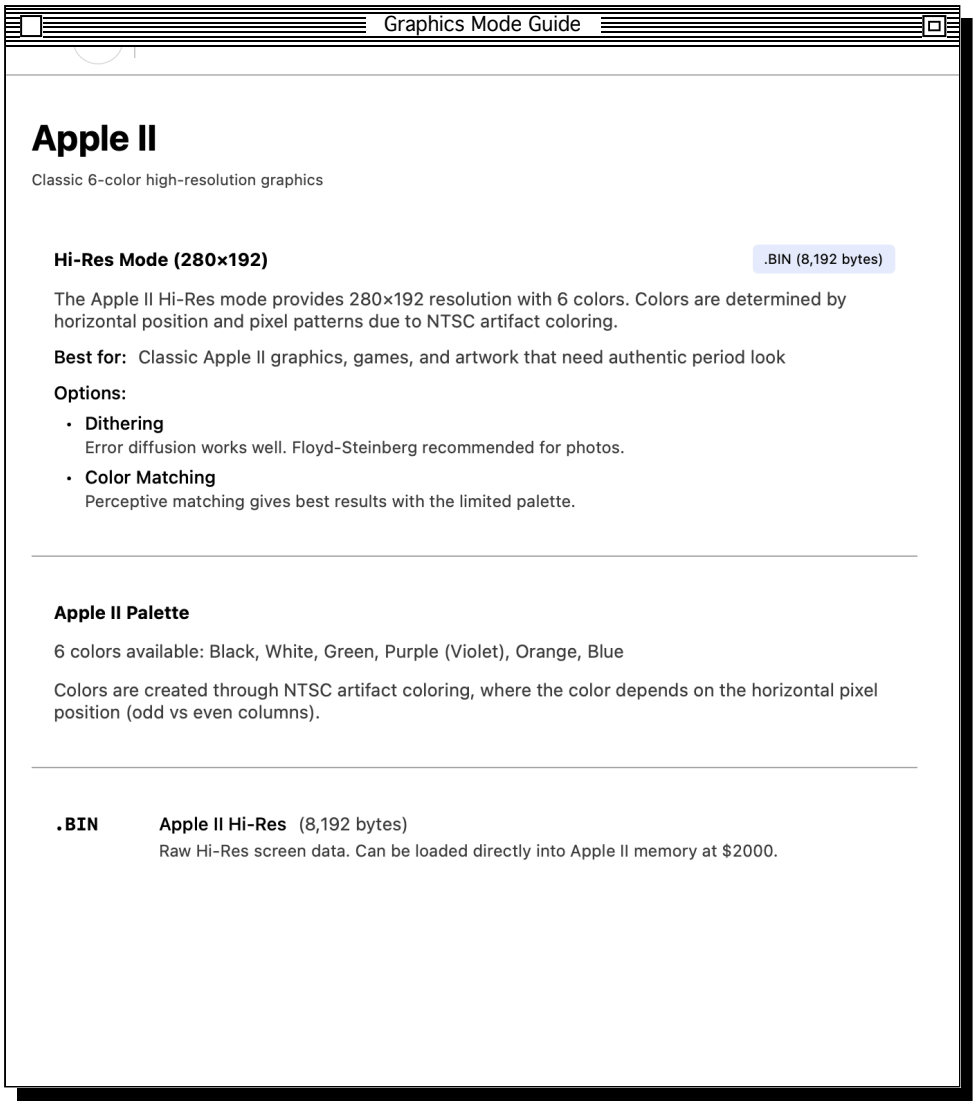


FIGURE 10-3
The Graphics Mode Guide (the system list on the left is not shown).

The Graphics Mode Guide and this guide cover the same ground. The Guide is always at hand inside the app; this one adds pictures, comparisons and the file formats.

CHAPTER 11

11 Questions and Answers

Problems and how to solve them

Short answers to the questions people ask most often.

The picture looks nothing like the original.

That is expected with the smallest palettes. Try more contrast (CLAHE), a little more saturation, or a different colour match. Pictures with strong shapes convert better than detailed ones. Chapter 4 has tips for choosing pictures.

There are black bars around the picture.

Fit is set to Contain, which shows the whole picture. Set Fit to Stretch to fill the screen, or to Cover to fill it and crop the edges.

A slider does nothing.

The selected picture is probably locked (the padlock is orange). A locked picture keeps its own settings. Click the padlock to release it.

My palette edits disappeared.

Any other change converts the picture again and discards palette edits. Make them last, after every other setting is final (Chapter 7).

An Atari 800 picture in Graphics 15 is nearly grey.

Four colours from 128 have to cover the whole picture. A picture with a lot of grey (a pale sky, a dark mountain) gets grey colours. Try more saturation, or a picture with more colour.

The IIgs 640 pictures look very wide.

Their pixels are twice as tall as they are wide, so the preview shows them at double height to look right. They are shown as they would be on the machine.

macOS says BitPast cannot be opened.

This can happen with a copy downloaded from GitHub. Control-click the app, choose Open, and confirm (Chapter 2).

A picture is missing from my disk image.

It did not fit. Choose a larger disk size and create the disk again.

The preview takes a moment to update.

Some conversions — HAM, the IIgs 3,200-colour mode and Per-Line SCB mode — are heavier than others. BitPast works in the background, so you can keep using the window; a Converting… indicator shows while it works.

The same picture converts differently each time.

Only the Noise dither is random. Every other setting gives exactly the same result for the same picture and settings.

CHAPTER A

A Native File Formats

Every file, its size and its layout

Every mode and the file BitPast writes for it. Sizes were measured with BitPast 4.6. Files that are compressed — IFF, PCX and ANSI — vary with the picture.

SYSTEM	MODE	FILE	SIZE
Apple II	DHGR (Double Hi-Res)	`.A2FC`	16,384 bytes
	HGR (Hi-Res)	`.BIN`	8,192 bytes
	LGR (Lo-Res)	`.SLO`	962 bytes
	DLGR (Double Lo-Res)	`.DLO`	1,922 bytes
	Mono (Monochrome)	`.A2FM`	16,384 bytes
	Mono, 280×192	`.BIN`	8,192 bytes
Apple IIgs	3200 Colors (Brooks)	`.3200`	38,400 bytes
	256 Colors (16 Palettes)	`.SHR`	32,768 bytes
	320x200 (16 Colors)	`.SHR`	32,768 bytes
	640x200 (4 Colors)	`.SHR`	32,768 bytes
	640x200 Enhanced (16 Colors)	`.SHR`	32,768 bytes
	640x200 Desktop (16 Colors)	`.SHR`	32,768 bytes
Amiga 500	Standard (32 colors)	`.iff`	varies
	HAM6 (4096 colors)	`.iff`	varies
Amiga 1200	Standard (256 colors)	`.iff`	varies
	HAM8 (262144 colors)	`.iff`	varies
Amstrad CPC	Mode 1 (320×200, 4 colors)	`.scr`	16,512 bytes
	Mode 0 (160×200, 16 colors)	`.scr`	16,512 bytes
Atari 800	Graphics 8 (320×192, 2 colors)	`.gr8`	7,682 bytes
	Graphics 15 (160×192, 4 colors)	`.gr15`	7,684 bytes
	Graphics 9 (80×192, 16 shades)	`.gr9`	7,696 bytes
	Graphics 10 (80×192, 9 colors)	`.gr10`	7,689 bytes
	Graphics 11 (80×192, 16 hues)	`.gr11`	7,696 bytes
Atari ST	Low Res (320×200)	`.pi1`	32,034 bytes
	Medium Res (640×200)	`.pi2`	32,034 bytes
	High Res (640×400)	`.pi3`	32,034 bytes
BBC Micro	Mode 0 (640×256, 2 colors)	`.bbc`	20,483 bytes
	Mode 1 (320×256, 4 colors)	`.bbc`	20,485 bytes
	Mode 2 (160×256, 8 colors)	`.bbc`	20,489 bytes
	Mode 4 (320×256, 2 colors)	`.bbc`	10,243 bytes
	Mode 5 (160×256, 4 colors)	`.bbc`	10,245 bytes
Commodore 64	HiRes (320×200)	`.art`	9,002 bytes
	Multicolor (160×200)	`.kla`	10,003 bytes
	PETSCII (40×25)	`.prg`	varies
MSX	Screen 2 (MSX1)	`.sc2`	13,063 bytes

SYSTEM	MODE	FILE	SIZE
	Screen 5 (MSX2)	.sc5	27,175 bytes
	Screen 8 (MSX2)	.sc8	54,279 bytes
IBM PC	CGA (4 colors)	.pcx	varies
	EGA (16 colors)	.pcx	varies
	EGA 64 (16 from 64)	.pcx	varies
	VGA (256 colors)	.pcx	varies
	CGA 80×25 Text	.ans	varies
	VESA 132×50 Text	.ans	varies
Commodore Plus/4	HiRes (320×200)	.prg	10,000 bytes
	Multicolor (160×200)	.prg	10,002 bytes
TRS-80 Color Computer	PMODE 3 (128×192, 4 colors)	.bin	6,144 bytes
	PMODE 4 (256×192, 2 colors)	.bin	6,144 bytes
	PMODE 1 (128×96, 4 colors)	.bin	3,072 bytes
	PMODE 2 (128×192, 2 colors)	.bin	3,072 bytes
	CoCo 3 (320×200, 16 colors)	.cm3	32,000 bytes
	CoCo 3 (640×200, 4 colors)	.cm3	32,000 bytes
VIC-20	HiRes (176×184)	.prg	varies
	LowRes (88×184)	.prg	varies
ZX Spectrum	Standard (256×192)	.scr	6,912 bytes

Layouts of some files

FILE	LAYOUT
.shr (IIgs)	32,768 bytes: 32,000 bytes of bitmap (200 lines of 160 bytes), 200 scan control bytes at $7D00, and 16 palettes of 16 colours at $7E00. Each colour is a 16-bit word, $0RGB, with four bits per component.
.3200 (IIgs, Brooks)	38,400 bytes: 32,000 bytes of bitmap, then a palette of 16 colours for each of the 200 lines (32 bytes each), with the colours stored from 15 down to 0.
.scr (ZX Spectrum)	6,912 bytes: 6,144 bytes of bitmap in the Spectrum's interleaved order, then 768 attribute bytes (ink, paper, brightness).
.scr (Amstrad CPC)	16,512 bytes: a 128-byte AMSDOS header followed by 16,384 bytes of screen memory.
.pi1 .pi2 .pi3 (Atari ST)	32,034 bytes: a 2-byte resolution word, the 16-word palette, and 32,000 bytes of bitplane data (DEGAS Elite format).
.kla (C64)	10,003 bytes: a 2-byte load address, 8,000 bytes of bitmap, 1,000 bytes of screen colours, 1,000 bytes of colour memory and the background colour (Koala Painter format).
.iff (Amiga)	An IFF ILBM picture: the standard Amiga picture format, with the palette and bitplanes (and the HAM flag in HAM modes).
.pcx (PC)	A PC Paintbrush picture with run-length compression; the palette is stored in the file.

CHAPTER B

B Colour Palettes

The palettes of the fixed-palette systems

The fixed palettes that BitPast chooses from, as it defines them. Systems whose palettes are chosen per picture from a very large set (the Amiga, the Atari ST, the Apple IIgs) are not shown.

Commodore 64 · 16 colours

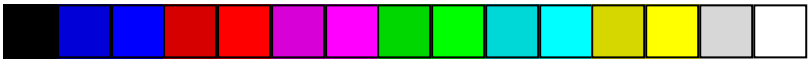


The Pepto palette.

VIC-20 · 16 colours

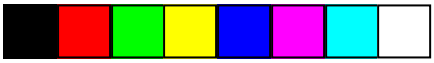


ZX Spectrum · 15 colours

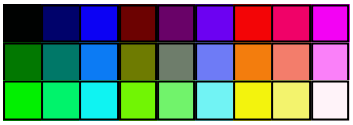


Eight colours at normal brightness and seven at bright (bright black is black).

BBC Micro · 8 colours



Amstrad CPC · 27 colours



Three levels each of red, green and blue.

MSX1 (TMS9918) · 16 colours

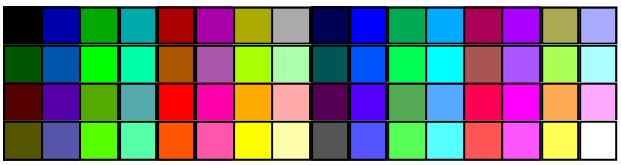


Index 0 is transparent and is shown as black.

PC CGA · 16 colours



PC EGA · 64 colours



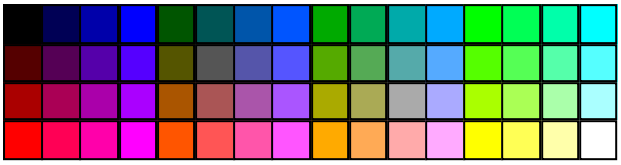
Two bits each of red, green and blue.

Apple IIgs Desktop · 16 colours



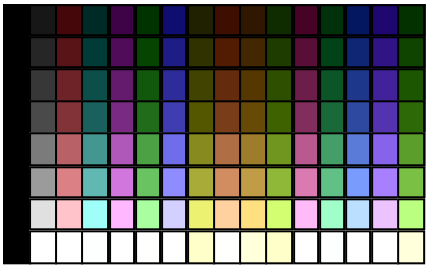
The palette of the 640×200 Desktop mode (even and odd columns share these colours in two groups of four).

TRS-80 CoCo 3 · 64 colours



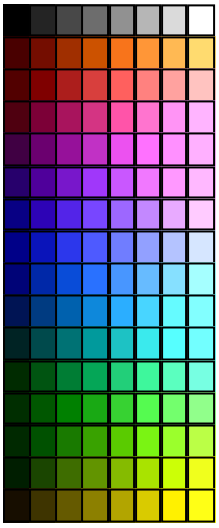
Two bits each of red, green and blue.

Commodore Plus/4 · 128 colours



Eight rows of brightness, from dark to light, each with the sixteen hues of the TED chip. Hue 0 is black in every row; the lightest row is almost white.

Atari 800 · 128 colours



Sixteen hues, one to a row, each in eight levels of brightness. The first row is grey.

CHAPTER C

C Glossary

Words used in this guide

Words used in this guide.

Artifact colour — Colour produced not by the video chip but by the way a television decodes patterns of dots. The Apple II and the CoCo's PMODE 4 rely on it.

Attribute clash — The look of the ZX Spectrum's colours: each 8×8 cell has only two colours, so a third colour in a cell makes it take on the colours of its neighbours.

Bitplane — One layer of a picture: the bits of the same weight for every pixel. Amiga and Atari ST pictures are stored as several bitplanes.

Brooks format — The file format for Apple IIgs pictures that have a full palette for each of the 200 lines, 3,200 colours in all.

Cell — A small block of the screen, such as 8×8 pixels, that has its own colours. Most 8-bit systems restrict the colours per cell.

Dithering — Mixing dots of the available colours so that the eye sees a colour that isn't in the palette.

Error diffusion — A dither that spreads the rounding error of each pixel over the pixels that are still to come.

HAM — Hold-And-Modify, an Amiga mode in which a pixel either takes a palette colour or changes one component of the pixel before it, giving thousands of colours.

Interlace — Showing a picture in two passes of alternate lines, which doubles the vertical resolution of the Amiga's screen.

Native file — The file a machine loads: the screen memory as the hardware stores it.

Ordered dithering — A dither that compares each pixel with a fixed pattern of thresholds, such as a Bayer matrix.

Palette — The set of colours a picture may use. Some machines have one palette, some many.

PETSCII — The Commodore character set, which includes graphic symbols. PETSCII pictures are made of characters rather than pixels.

SCB — Scan Control Byte: on the Apple IIgs, a byte for each line that chooses its palette and mode.

Super Hi-Res — The Apple IIgs's graphics screen: 320×200 or 640×200 pixels with palettes of 16 colours from 4,096.

Volume name — The name of a disk. Every system limits its length and characters.

CHAPTER D

D Credits and Licences

Thanks, sources and licences

BitPast is the work of Walter Tengler, with the help of the people and projects below.

b2d

The Apple II conversions are made by b2d (BMP2DHR) by Bill Buckels, built into BitPast with changes for modern macOS. See appleoldies.ca/bmp2dhr.

Fonts

The retro appearances use the Print Char 21 and Shaston bitmap fonts included with BitPast.

Photographs

The sample photographs in this guide are from Pixabay and are used under the Pixabay Content License: the bee-eater by davidclode, Nyhavn in Copenhagen by ductinh91, the hot-air balloon by guojinliang4104, the toucan by hbieser, the carnival dancer by makabera, the colour pencils by margaret8, the popsicles by motion_vfx_tlx, the flowers by naturerelax and the rainbow by sergiovisor_ph.

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Licence

BitPast's user interface code is provided under the MIT License. The source code is at github.com/portwally/BitPast.

This guide

Set in Apple Garamond, Myriad-style headings and Geneva, in the manner of the printed guides of the early 1990s. All pictures of converted graphics, and all screenshots, were made with BitPast version 4.6.